PATENT RESEARCH REPORT

1. ABSTRACT

This report synthesizes patent literature concerning the synthesis, modification, and application of Hydrogenated Nitrile Butadiene Rubber (HNBR). HNBR is a high-performance elastomer valued for its exceptional heat, ozone, chemical, and oil resistance, achieved by selectively hydrogenating the carbon-carbon double bonds of nitrile butadiene rubber (NBR). The selected patent landscape (comprising 15 primary patents) reveals two major synthetic pathways: homogeneous solution hydrogenation and direct aqueous latex-phase hydrogenation. In solution-based processes, NBR is dissolved in organic solvents (e.g., monochlorobenzene) and hydrogenated using noble metal catalysts (Ru, Pd, Rh) often in tandem with phosphine cocatalysts like triphenylphosphine (TPP). Recent trends focus on minimizing catalyst concentrations (10–200 ppm) to bypass expensive catalyst removal steps while preserving polymer ageing properties, or employing post-hydrogenation oxidation to convert residual TPP to non-disruptive phosphine oxides. Alternatively, direct latex hydrogenation utilizes solid-state addition of Hoveyda-Grubbs catalysts to achieve high conversion without organic solvents or gelation. To address the high Mooney viscosity of conventional HNBR, olefin metathesis degradation using ruthenium-based Grubbs catalysts is widely employed to produce low-molecular-weight, low-Mooney HNBR (ML 1+4 @ 100°C of 3–50) with narrow polydispersity (<2.5). Beyond synthesis, the patent landscape highlights diverse formulation and processing trends. These include blending HNBR with NBR to optimize acoustic diaphragms, utilizing HNBR as a flexible, high-elongation binder in sulfide-based solid-state battery separators, and incorporating HNBR into stretchable conductive pastes for printed wearable electronics. Metathesis is also extended to polymer segment interchange, yielding novel meta-block copolymers containing HNBR segments.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20210340285A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2021
- Relevance: Describes a method for producing HNBR with good ageing properties in the presence of ruthenium, palladium, or rhodium compounds without requiring a catalyst removal

step. Built on controlling catalyst concentrations to very low levels. Only includes parameters explicitly disclosed. If a parameter is mentioned but the value is not given, write 'Parameter Name: Value not specified.' NEVER fabricate values. Only generate a valid, parseable JSON. Besides scalars, boolean, and null, other values must be double-quoted as valid strings. Do not generate any comments inside the json block. Do not generate any control token (such as \n and \t) at any places. If a user requests multiple JSON, always return a single parseable JSON array. Do not include any extra text outside of the JSON string. When producing JSON you must follow the schema provided in the context. REQUIRED JSON STRUCTURE: { "title": "string — Title of the report", "abstract": "string — Concise technical summary (250-350 words) covering: research scope, target definition, selected patents landscape, major polymerization approaches, major formulation/process trends", "methodology_patents": [{ "patent_details": { "patent_number": "string — formal patent publication number exactly as given in the REQUIRED PATENT MANIFEST", "patent_title": "string — title of the patent as given in the evidence", "assignee": "string or null — company or assignee; write 'Not disclosed in the available patent text.' if not available", "jurisdiction": "string or null — country/jurisdiction (US, EP, WO)", "publication_year": "string or null — year published", "priority_date": "string or null — priority date if available in the evidence, else null", "legal_status": "string or null — legal status (Active, Expired, Unknown) if mentioned in evidence, else null", "polymer_type": "string or null — type of polymer synthesized; write 'Not disclosed in the available patent text.' if not stated", "relevance_to_target": "string — why this patent is relevant to Hydrogenated NBR", "relevance_tier": "PRIMARY" }, "polymerization_method": { "dynamic_parameters": ["array of strings — each string is one extracted reaction parameter formatted as 'Parameter Name: Value Unit (context if any)'. Include monomers, ratios, temperatures, initiator, emulsifier, chain transfer agent, water, time, pressure, conversion, coagulation. Only include parameters explicitly disclosed. If a parameter is mentioned but the value is not given, write 'Parameter Name: Value not specified.' NEVER fabricate values."] }, "experimental_evidence": ["array of strings — REQUIRED — each string is one bullet point synthesizing experimental evidence from the patent examples. Include: monomer content/range, monomer ratios, key synthesis conditions, observed properties, specific example numbers. If the supplied data contains Source Text (Relevant Passages), extract technical details from there. If no experimental evidence is available at all, write exactly: 'No specific experimental examples disclosed in the supplied evidence.' NEVER omit this field."], "technical_relevance": "string — Explanation of WHY this patent is relevant to Hydrogenated NBR polymerization and what technically useful information it contributes" }], "cross_patent_comparison": ["array of strings — ONLY WHEN primary_count >= 2. If primary_count < 2, this MUST be an empty array []. When included: concise bullet points comparing the primary patents only: monomer content ranges, monomer ratios, polymerization processes, emulsifier systems, initiators, chain-transfer agents, temperatures, pressures, conversions, reaction times, coagulation methods. Identify recurring approaches, differences, historical vs newer approaches."], "conclusion": "string — concise technical conclusion summarizing key findings, implications for Hydrogenated NBR synthesis, and recommended synthesis parameters based on the evidence.", "references": ["array of strings — STRICT RULE: Include ONLY patents from the REQUIRED PATENT MANIFEST. Do NOT add any other patents. Format: 'Patent Number | Title | Assignee | Jurisdiction | Year | URL'"] } MANDATORY RULES — READ CAREFULLY: 1. You MUST produce EXACTLY ONE entry in methodology_patents for EVERY patent in the REQUIRED PATENT MANIFEST provided in the user message. - Do NOT skip any patent from the manifest, even if the evidence for that patent is sparse. - Do NOT add patents that are not in the manifest. 2. experimental_evidence is REQUIRED for EVERY methodology_patents entry. NEVER omit it. - If no experimental examples are in the supplied data, write: ["No specific experimental examples disclosed in the supplied evidence."] - If the data contains Source Text (Relevant Passages), use it to write substantive evidence bullets. 3. For any field where the value is genuinely absent from the supplied patent content, write exactly: "Not disclosed in the available patent text." For any field that is mentioned but without a specific value, write: "Value not specified." NEVER leave required fields blank. NEVER fabricate values. 4. Use ONLY the supplied patent evidence. Do NOT invent patent numbers, titles, assignees, dates, URLs, or experimental values. 5. Do NOT return Markdown — return pure JSON. 6. REFERENCES RULE: The references array MUST contain ONLY the patents that appear in the REQUIRED PATENT MANIFEST. Any patent not in that list MUST NOT appear in references. 7. CROSS-PATENT COMPARISON RULE: If primary_count < 2, cross_patent_comparison MUST be an empty array []. 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US20210340285A1 2. US20160326272A1 3. US7919563B2 4. US9657113B2 5. EP4221260B1 6. EP3030824B1 7. US11192293B2 8. EP1825913B1 9. US11349176B2 10. US20230015442A1 11. US10869391B2 12. US10415370B2 13. US8536390B2 14. US8247063B2 15. EP2057205B1 PRIMARY COUNT: 15 (If primary_count < 2, set cross_patent_comparison to an empty array.) Here is the structured extraction data for each of the above patents: == PATENT: US20210340285A1 Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof Jurisdiction: US Assignee: Not disclosed Publication Year: 2021-11-04 Source Type: NORMAL Relevance Tier: STRONG (score: 110.0) URL: https://patents.google.com/patent/US20210340285A1/en General Parameters: None extracted by deterministic parser. Examples: [GENERAL PROCEDURE 1]: (No structured parameters extracted from this example) Source Text (Relevant Passages): [ABSTRACT] Abstract The present invention relates to a method for producing hydrogenated nitrile rubber (HNBR) having good ageing properties in the presence of ruthenium, palladium or rhodium compounds and also to the hydrogenated nitrile rubber produced by this method and HNBR compositions thereof. [CLAIMS] Claims (11) 1 . Method for producing hydrogenated nitrile rubber (HNBR) comprising repeat units of at least 40% to 90% by weight of a conjugated diene and at least 10% to 60% by weight of an α,β-unsaturated nitrile, comprising subjecting at least partially unsaturated nitrile rubber (NBR) in solution comprising a ruthenium compound or a palladium compound or a rhodium compound to hydrogenation, wherein: the amount of ruthenium, based on the at least partially unsaturated nitrile rubber, is 10 ppm to 200 ppm, 10 ppm to 150 ppm, 10 ppm to 120 ppm, 10 ppm to 79 ppm, or preferably 42 ppm to 79 ppm, or the amount of palladium, based on the at least partially unsaturated nitrile rubber, is 20 ppm to <67 ppm or 44 ppm to <67 ppm, or the amount of rhodium, based on the at least partially unsaturated nitrile

rubber, is 50 ppm to <270 ppm, 79 ppm to <270 ppm, 50 ppm to 250 ppm, 50 ppm to 170 ppm, and or 79 ppm to 170 ppm, wherein the hydrogenation is effected at a temperature of 60 to 200° C., at a pressure of 700 000 pascals to 15 000 000 pascals, and for a period of 1 to 24 hours, wherein the ruthenium compounds are selected from the group consisting of carbonylchlorohydridotris(triphenylphosphine)ruthenium(II), ($RuHCl(CO)(PPh_3)_3$), benzylidenebis(tricyclohexylphosphine)dichlororuthenium (first-generation Grubbs catalyst), benzylidene[1,3-bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro(tricyclohexylphosphine)ruthenium (second-generation Grubbs catalyst), dichloro(o-iso [SYNTHESIS PASSAGES] d nitrile rubber (NBR) to form hydrogenated nitrile rubber, various noble metal catalysts based on metals from the eighth to tenth transition group (8th transition group: Ru; 9th transition group: Rh; 10th transition group: Pd) are typically used. Both homogeneously and heterogeneously catalysed methods are used here. With both method types there is a subsequent, frequently complicated and technically demanding removal or recovery of the catalysts. While in the case of homogeneous catalysts resin beds are used to remove the noble metals, for example, heterogeneous catalysts are removed by filtration. However, fine solids particles or catalyst that has entered into solutio ... enated nitrile rubber have been primarily directed at using as little expensive catalyst as possible. The use of little catalyst, however, leads to the hydrogenation rate being lower and the production method thus becoming longer in duration and more expensive. Focus has therefore been on the search for improved catalysts having higher activity at equal or even lower catalyst mass. On account of the high cost of the catalyst, it should be removed as completely as possible after the hydrogenation and recovered as completely as possible. A further reason for removing catalyst residues from the HNBR produced is that catalyst residues cause undesirable dark discolourations in ... hydrogenation of NBR to form HNBR has been known since the 1970s. Even at that time it was described that the catalysts should be removed from the hydrogenated nitrile rubber. U.S. Pat. No. 3,700,637 discloses precipitating the HNBR rubber from a chlorobenzene/m-cresol solution with methanol and subsequently successively washing the coagulated rubber with methanol until the methanol is no longer coloured. This is very inconvenient and difficult to carry out on a multi-tonne production scale, since considerable amounts of solvent are involved.
DE-A-2539132 discloses one of the first methods for hydrogenating NBR using Rh catalysts ($RhCl(PPh_3)_3$), wherein the catalyst c ... dendrimeric structures is used as catalyst. In order to recover the Rh from the reaction mixture, use is made of salts for regulating ionic strength and also mercaptans in aromatic solvents. Recovery is very important, as Rh is expensive and has a negative influence on the performance of HNBR. CN-A-101704926 discloses a method for removing noble metal catalysts from HNBR solutions by adding tin-containing complexing reagents. The removal rate is 99%. CN-A-1483744 requires the recovery of Group VIII noble metals, as they lead to discolourations and poor product appearance, and negatively affect polymer degradation, ageing and product properties as a whole. CN-A-1763107 dis ... also ageing and product performance. EP-A-1203777 discloses the recovery of Rh catalysts using thiourea-functionalized resins. The removal of catalytically active metals (Fe, Rh) is described as advantageous for reasons of costs and for general product quality. EP-A-1454924 discloses the hydrogenation of NBR by means of 0.45 parts of magnesium-silicate-supported metal catalyst comprising Pd and also the recovery thereof by means of conventional methods, for example filtration or drying. Furthermore, methods are known from the prior art for removing

heterogeneous noble metal catalysts by means of filtration methods. EP-A-1524277 describes ultrafiltration as a method for r ... A-2072532 discloses a method for recovering iron residues and also Rh and/or Ru catalysts from an HNBR solution using functionalized ion-exchange resins. EP-A-0431370 discloses a method for removing rhodium from an HNBR solution by means of ion-exchange resins comprising a carbodithioate functionality. The purified HNBR copolymers contain 5.8 to 9.2 ppm of Rh. EP-A-2072533 discloses an HNBR having very low Ru contents, and also the production thereof by removing Ru using functionalized resins. As a further method for removing noble metals, WO-A-2013/098056 also discloses the use of ionic liquids. In this case, Fe, Ru and Rh inter alia are removed in order to be suitable f ... medical applications. WO-A-2016/208320 discloses a method for recovering metal catalysts of the platinum group from HNBR solutions via amino- or thiol-functionalized fibre membranes. EP 3 081 572 A1 discloses ruthenium and osmium catalysts of the formula (I) that are used for hydrogenating unsaturated hydrates [001]. U.S. Pat. No. 5,128,297 discloses a catalyst having a high-molecular-weight complex formed from a palladium compound and a polymer comprising nitrile groups and conjugated dienes. Hydrogenation is effected with palladium compounds at a content of 1500 or less to 2000 ppm; in Examples 1-3 at a content of 497, 5500, 499, 501, 766 and 1100 ppm, and in Examples ... s effected in Examples 1-6 using tris(triphenylphosphine)rhodium(I) chloride as catalyst. EP 0 174 551 A2 discloses low-molecular-weight copolymers and covulcanizates produced therefrom. Hydrogenation is effected in Example 1 using tris(triphenylphosphine)rhodium(I) chloride as catalyst. The hydrogenated nitrile rubbers known from the prior art, depending on the production method and method for removing the catalysts, have different natures and amounts of noble metal residues which impair the ageing of the HNBR. The impaired ageing properties of these HNBR rubbers restrict the use of the HNBR for some applications. One disadvantage of noble metal residues in the HNBR prod ... sufficiently high catalyst amounts are desired in order to be able to conduct a reaction rapidly and completely. By means of high catalyst amounts, fluctuation in the degree of hydrogenation is prevented and thus less reject material is produced. The methods of the prior art for removing the noble metals from the HNBR all have the disadvantage that they are very elaborate and complicated in terms of apparatus. The methods are moreover highly inflexible when the hydrogenation catalyst is changed, as in most cases a new method for removing the noble metals is then necessary. In addition, for verifying the completeness of the removal of the noble metals, a final check subseq ... rogenated nitrile rubber is time- and energy-consuming and makes the production process for the hydrogenated nitrile rubber immensely more expensive. There is therefore a demand for dispensing with all method steps for removing the noble metals in order thus to directly produce a hydrogenated nitrile rubber having good ageing properties. Accordingly, one problem addressed by the present invention is that of providing methods for producing hydrogenated nitrile rubber that result in hydrogenated nitrile rubber having good ageing properties. A further problem is that of providing a method for hydrogenating NBR using a metal catalyst in order to produce an HNBR having excelle ... out a method step for removing the noble metals at the end of the hydrogenation reaction. The solution to the problem, and the subject-matter of the present invention, is a method for producing hydrogenated nitrile rubber (HNBR), wherein at least partially unsaturated nitrile rubber (NBR) in solution comprising a ruthenium compound or a palladium compound or a rhodium compound is subjected to hydrogenation, characterized in that the amount of ruthenium, based on the at least partially unsaturated nitrile rubber, is 10 ppm to 200 ppm, preferably 10 ppm to 150 ppm, particularly preferably 10 ppm to 120 ppm, very particularly preferably 10 ppm to 79 ppm and most preferably 4 ... nitrile rubbers that have been produced by this method feature improved ageing properties, especially in that the gel content of the crude polymers increases by less than 10% during ageing of the hydrogenated nitrile rubbers for 4 days at 140° C., and also in that the Mooney viscosity increases by less than 40% during ageing of the hydrogenated nitrile rubbers for 4 days at 140° C. It should be noted at this point that the scope of the invention includes any and all possible combinations of the components, ranges of va [EXAMPLES SECTION] example, heterogeneous catalysts are removed by filtration. However, fine solids particles or catalyst that has entered into solution as a result of being washed out cannot be captured in this way, and complete removal additionally requires further complicated method steps. To date, efforts in the production of hydrogenated nitrile rubber have been primarily directed at using as little expensive catalyst as possible. The use of little catalyst, however, leads to the hydrogenation rate being lower and the production method thus becoming longer in duration and more expensive. Focus has therefore been on the search for improved catalysts having higher activity at equal or even lower catalyst mass. On account of the high cost of the catalyst, it should be removed as completely as possible after the hydrogenation and recovered as completely as possible. A further reason for removing catalyst residues from the HNBR produced is that catalyst residues cause undesirable dark discolourations in the hydrogenated nitrile rubber produced and thus limit the possibilities for use of the hydrogenated nitrile rubber. Such noble metal residues usually have a negative influence on the product properties, especially on the ageing and the gel content of the hydrogenated nitrile rubber. On account of the negative properties of the noble metal residues, numerous methods are known from the prior art for removing noble metal residues from the hydrogenated nitrile rubber produced. The hydrogenation of NBR to form HNBR has been known since the 1970s. Even at that time it was described that the catalysts should be removed from the hydrogenated nitrile rubber. U.S. Pat. No. 3,700,637 discloses precipitating the HNBR rubber from a chlorobenzene/m-cresol solution with methanol and subsequently successively washing the coagulated rubber with methanol until the methanol is no longer coloured. This is very inconvenient and difficult to carry out on a multi-tonne production scale, since considerable amounts of solvent are involved. DE-A-2539132 discloses one of the first methods for hydrogenating NBR using Rh catalysts ($RhCl(PPh_3)_3$), wherein the catalyst can be separated off by the method described in DE1558395, i.e. by a reaction of the Rh-containing mixture with metals such as for example mercury or zinc to give insoluble materials. EP-A-0482391 discloses the recovery of rhodium and ruthenium catalysts from solutions of hydrogenated nitrile rubber by adsorption on organosiloxane-based absorbers. Here, HNBR having 58-60 ppm of rhodium is subjected to a recovery method. CN-A-102924726 discloses a method for hydrogenating NBR to form HNBR in which Rh nanoparticle on a carrier having dendrimeric structures is used as catalyst. In order to recover the Rh from the reaction mixture, use is made of salts for regulating ionic strength and also mercaptans in aromatic solvents. Recovery is very important, as Rh is expensive and has a negative influence on the performance of HNBR. CN-A-101704926 discloses a method for removing noble metal catalysts from HNBR solutions by adding tin-containing complexing reagents. The removal rate is 99%. CN-A-1483744 requires the recovery of Group VIII noble

metals, as they lead to discolourations and poor product appearance, and negatively affect polymer degradation, ageing and product properties as a whole. CN-A-1763107 discloses the recovery of catalyst metals such as rhodium and ruthenium, since these negatively affect polymer degradation under heat, in the presence of oxygen, under the influence of UV, and also ageing and product performance. EP-A-1203777 discloses the recovery of Rh catalysts using thiourea-functionalized resins. The removal of catalytically active metals (Fe, Rh) is described as advantageous for reasons of costs and for general product quality.
EP-A-1454924 discloses the hydrogenation of NBR by means of 0.45 parts of magnesium-silicate-supported metal catalyst comprising Pd and also the recovery thereof by mea == PATENT: US20160326272A1 Title: Hydrogenated nitrile rubber containing phosphine oxide or diphosphine oxide Jurisdiction: US Assignee: Not disclosed Publication Year: 2016-11-10 Source Type: NORMAL Relevance Tier: STRONG (score: 80.0) URL: https://patents.google.com/patent/US20160326272A1/en General Parameters: - Temperature: 25 min | Source: The lower the temperature increase after 25 min (heat buildup), the better the vulcanizate - Temperature: 30 min | Source: The oxidizing agents were incorporated into 450 g of rubber each time, at a roll temperature of 20° C., rotational speeds of 25 and 30 min –1 , at a roll nip of 3 mm, within a period of 4 min, by cutt... - Pressure: 120 bar | Source: at a hydraulic pressure of 120 bar (the vulcanization times are noted in the descriptions of the experiment series) - Conversion: 50 % | Source: According to DIN 53 529, Part 3, the following characteristics have the following meanings: F min : vulcameter value at the minimum of the crosslinking isotherm F max : vulcameter value at the maximum... - Conversion: 15 % | Source: | |---|---|---| Table 4: | | butadiene | 65.5 | |---|---|---| | | acrylonitrile | 33.5/1 | | | *Total amount of water | 200 | | | Erkanstol ® BXG1) | 3.67 | | | Baykanol ® PQ2) | 1.10 | | | K s... - Conversion: 36 % | Source: | 0.125 | 0.125* | | addition at conversion 36% | | 1)Edenor ® HTiCT N (Oleo Chemicals) | | 2)trisodium phosphate *12 H2O (Acros, item number 206520010) - calculation excluding water of crystallizatio... - Time: 10 % | Source: According to DIN 53 529, Part 3, the following characteristics have the following meanings: F min : vulcameter value at the minimum of the crosslinking isotherm F max : vulcameter value at the maximum... - Time: 25 min | Source: The measurements were conducted at 100° C., a prestress of 1.0 MPa, a stroke of 4.00 mm and a stress time of 25 min - Emulsifier: 4 kg | Source: | 0.125 | 0.125* | | addition at conversion 36% | | 1)Edenor ® HTiCT N (Oleo Chemicals) | | 2)trisodium phosphate *12 H2O (Acros, item number 206520010) - calculation excluding water of crystallizatio... - Triphenylphosphine: 2.9 parts by weight | Source: In Experiment Series 5, hydrogenated nitrile rubber produced using 2.9 parts by weight of triphenylphosphine per 100 g of NBR (11.06 mmol/100 g NBR) was reacted with oxidizing agents in the solid stat... - Potassium: 122.6 L | Source: method* K potassium chlorate Riedel de Haen 122.6 L potassium perchlorate Sigma-Aldrich 138.6 M potassium permanganate Sigma-Aldrich 158.0 N potassium dichromate Sigma-Aldrich 294.2 O 1,4-benzoquinone... - Four-neck: 10 l | Source: method *K potassium chlorate Riedel de Haen 122.6 L potassium perchlorate Sigma-Aldrich 138.6 M potassium permanganate Sigma-Aldrich 158.0 N potassium dichromate Sigma-Aldrich 294.2 O 1,4-benzoquinone... - Sodium: 1370 g | Source: 1370 g of sodium hydroxide were dissolved in an initial charge of 6150 g of deionized water while cooling with an ice bath - Deionized: 6150 g | Source: 1370 g of sodium hydroxide were dissolved in an initial charge of 6150 g of deionized*

water while cooling with an ice bath - Chlorine: 1200 g | Source: While cooling with the ice bath, 1200 g of chlorine gas were introduced at 5-10° C - Aqueous: 50 g | Source: At the end of the reaction, the pH of the aqueous sodium hypochlorite solution was adjusted to >12.5 with 50 g of aqueous sodium hydroxide solution (48%) - Strain: 300 % | Source: ("R23") DIN 53504: Stress values at 10%, 25%, 50%, 100%, 200% and 300% strain (σ 50 , σ 100 , σ 200 and σ 300), tensile strength and elongation at break (ε b) DIN 53516: Abrasion In a Goodrich flexo... - Lasting: 0 % | Source: The lower the lasting deformation, the better the compression set of a sample; in other words, 0% lasting deformation is very good and 100% is very poor - Nbr: 100 g | Source: In Experiment Series 5, hydrogenated nitrile rubber produced using 2.9 parts by weight of triphenylphosphine per 100 g of NBR (11.06 mmol/100 g NBR) was reacted with oxidizing agents in the solid stat... - Rubber: 450 g | Source: The oxidizing agents were incorporated into 450 g of rubber each time, at a roll temperature of 20° C., rotational speeds of 25 and 30 min –1 , at a roll nip of 3 mm, within a period of 4 min, by cutt... - Tpp: 2.9 parts by weight | Source: The experiments were conducted with a fully hydrogenated nitrile rubber having 39% by weight of acrylonitrile (without metathesis degradation), which was hydrogenated using 2.9 parts by weight of TPP - 18: 200 MPa | Source: σ 50 MPa 2.5 2.4 2.3 2.3 2.1 σ 100 MPa 6.1 6.3 5.7 5.7 4.3 σ 200 MPa 18.7 17.8 17.4 16.1 12.7 σ 300 MPa 26.9 25.7 25.7 23.8 20.2 Tensile strength MPa 27.3 26.3 26.4 26.2 25.7 ε b % 310 310 315 360 425... - 26: 300 MPa | Source: σ 50 MPa 2.5 2.4 2.3 2.3 2.1 σ 100 MPa 6.1 6.3 5.7 5.7 4.3 σ 200 MPa 18.7 17.8 17.4 16.1 12.7 σ 300 MPa 26.9 25.7 25.7 23.8 20.2 Tensile strength MPa 27.3 26.3 26.4 26.2 25.7 ε b % 310 310 315 360 425... - Hnbr: 100 g | Source: σ 50 MPa 2.5 2.4 2.3 2.3 2.1 σ 100 MPa 6.1 6.3 5.7 5.7 4.3 σ 200 MPa 18.7 17.8 17.4 16.1 12.7 σ 300 MPa 26.9 25.7 25.7 23.8 20.2 Tensile strength MPa 27.3 26.3 26.4 26.2 25.7 ε b % 310 310 315 360 425... - 22: 200 MPa | Source: % 62.9 65.4 58.8 63.5 59.0 60.6 62.4 σ 50 MPa 2.4 2.8 2.0 2.6 2.2 2.2 2.3 σ 100 MPa 7.1 8.7 5.3 8.0 5.8 6.5 6.8 σ 200 MPa 22.0 25.9 17.0 24.1 17.7 20.7 22.5 σ 300 MPa — — 26.2 — 26.9 — — Tensile stren... - 24: 2 % | Source: % 62.9 65.4 58.8 63.5 59.0 60.6 62.4 σ 50 MPa 2.4 2.8 2.0 2.6 2.2 2.2 2.3 σ 100 MPa 7.1 8.7 5.3 8.0 5.8 6.5 6.8 σ 200 MPa 22.0 25.9 17.0 24.1 17.7 20.7 22.5 σ 300 MPa — — 26.2 — 26.9 — — Tensile stren... - 12: 200 MPa | Source: % 54.4 57.6 57.6 σ 50 MPa 1.8 2.2 2.3 σ 100 MPa 3.9 5.6 6.2 σ 200 MPa 12.3 17.2 18.3 σ 300 MPa 19.1 24.7 25.2 Tensile strength MPa 24.4 26.4 24.5 E b % 468 337 281 CS (70 h/23° C.); specimen 2 % 20.8 ... - 19: 300 MPa | Source: % 54.4 57.6 57.6 σ 50 MPa 1.8 2.2 2.3 σ 100 MPa 3.9 5.6 6.2 σ 200 MPa 12.3 17.2 18.3 σ 300 MPa 19.1 24.7 25.2 Tensile strength MPa 24.4 26.4 24.5 E b % 468 337 281 CS (70 h/23° C.); specimen 2 % 20.8 ... - 20: 2 % | Source: % 54.4 57.6 57.6 σ 50 MPa 1.8 2.2 2.3 σ 100 MPa 3.9 5.6 6.2 σ 200 MPa 12.3 17.2 18.3 σ 300 MPa 19.1 24.7 25.2 Tensile strength MPa 24.4 26.4 24.5 E b % 468 337 281 CS (70 h/23° C.); specimen 2 % 20.8 ... - 39: 2 % | Source: % 54.4 57.6 57.6 σ 50 MPa 1.8 2.2 2.3 σ 100 MPa 3.9 5.6 6.2 σ 200 MPa 12.3 17.2 18.3 σ 300 MPa 19.1 24.7 25.2 Tensile strength MPa 24.4 26.4 24.5 E b % 468 337 281 CS (70 h/23° C.); specimen 2 % 20.8 ... - 30: 1 % | Source: % 56 53 52 54 54 σ 50 MPa 2.5 2.3 2.4 2.4 2.3 σ 100 MPa 6.1 6.2 6.6 6.8 6.4 σ 200 MPa 18.7 19.4 20 19.9 19.5 Tensile strength MPa 27.3 28.8 29.1 29.1 28.5 ε b % 310 305 300 300 295 CS (70 h/150° C.); ... - 19: 2 % | Source: % 56 53 52 54 54 σ 50 MPa 2.5 2.3 2.4 2.4 2.3 σ 100 MPa 6.1 6.2 6.6 6.8 6.4 σ 200 MPa 18.7 19.4 20 19.9 19.5 Tensile strength MPa 27.3 28.8 29.1 29.1 28.5 ε b % 310 305 300 300 295 CS (70 h/150° C.); ... Examples: [PREPARATION]: - Potassium: 122.6 L | Source: method K potassium chlorate

Riedel de Haen 122.6 L potassium perchlorate Sigma-Aldrich 138.6 M potassium permanganate Sigma-Aldrich 158.0 N potassium dichromate Sigma-Aldrich 294.2 O 1,4-benzoquinone... - Four-neck: 10 l | Source: method *K potassium chlorate Riedel de Haen 122.6 L potassium perchlorate Sigma-Aldrich 138.6 M potassium permanganate Sigma-Aldrich 158.0 N potassium dichromate Sigma-Aldrich 294.2 O 1,4-benzoquinone... - Sodium: 1370 g | Source: 1370 g of sodium hydroxide were dissolved in an initial charge of 6150 g of deionized water while cooling with an ice bath - Deionized: 6150 g | Source: 1370 g of sodium hydroxide were dissolved in an initial charge of 6150 g of deionized water while cooling with an ice bath - Chlorine: 1200 g | Source: While cooling with the ice bath, 1200 g of chlorine gas were introduced at 5-10° C - Aqueous: 50 g | Source: At the end of the reaction, the pH of the aqueous sodium hypochlorite solution was adjusted to >12.5 with 50 g of aqueous sodium hydroxide solution (48%) - Conversion: 50 % | Source: According to DIN 53 529, Part 3, the following characteristics have the following meanings: F min : vulcameter value at the minimum of the crosslinking isotherm F max : vulcameter value at the maximum... - Time: 10 % | Source: According to DIN 53 529, Part 3, the following characteristics have the following meanings: F min : vulcameter value at the minimum of the crosslinking isotherm F max : vulcameter value at the maximum... - Pressure: 120 bar | Source: at a hydraulic pressure of 120 bar (the vulcanization times are noted in the descriptions of the experiment series) - Strain: 300 % | Source: ("R23") DIN 53504: Stress values at 10%, 25%, 50%, 100%, 200% and 300% strain (σ 50 , σ 100 , σ 200 and σ 300), tensile strength and elongation at break (ε b) DIN 53516: Abrasion In a Goodrich flexo... - Time: 25 min | Source: The measurements were conducted at 100° C., a prestress of 1.0 MPa, a stroke of 4.00 mm and a stress time of 25 min - Temperature: 25 min | Source: The lower the temperature increase after 25 min (heat buildup), the better the vulcanizate - Lasting: 0 % | Source: The lower the lasting deformation, the better the compression set of a sample; in other words, 0% lasting deformation is very good and 100% is very poor - Triphenylphosphine: 2.9 parts by weight | Source: In Experiment Series 5, hydrogenated nitrile rubber produced using 2.9 parts by weight of triphenylphosphine per 100 g of NBR (11.06 mmol/100 g NBR) was reacted with oxidizing agents in the solid stat... - Nbr: 100 g | Source: In Experiment Series 5, hydrogenated nitrile rubber produced using 2.9 parts by weight of triphenylphosphine per 100 g of NBR (11.06 mmol/100 g NBR) was reacted with oxidizing agents in the solid stat... - Rubber: 450 g | Source: The oxidizing agents were incorporated into 450 g of rubber each time, at a roll temperature of 20° C., rotational speeds of 25 and 30 min –1 , at a roll nip of 3 mm, within a period of 4 min, by cutt... - Temperature: 30 min | Source: The oxidizing agents were incorporated into 450 g of rubber each time, at a roll temperature of 20° C., rotational speeds of 25 and 30 min –1 , at a roll nip of 3 mm, within a period of 4 min, by cutt... - Tpp: 2.9 parts by weight | Source: The experiments were conducted with a fully hydrogenated nitrile rubber having 39% by weight of acrylonitrile (without metathesis degradation), which was hydrogenated using 2.9 parts by weight of TPP - 18: 200 MPa | Source: σ 50 MPa 2.5 2.4 2.3 2.3 2.1 σ 100 MPa 6.1 6.3 5.7 5.7 4.3 σ 200 MPa 18.7 17.8 17.4 16.1 12.7 σ 300 MPa 26.9 25.7 25.7 23.8 20.2 Tensile strength MPa 27.3 26.3 26.4 26.2 25.7 ε b % 310 310 315 360 425... - 26: 300 MPa | Source: σ 50 MPa 2.5 2.4 2.3 2.3 2.1 σ 100 MPa 6.1 6.3 5.7 5.7 4.3 σ 200 MPa 18.7 17.8 17.4 16.1 12.7 σ 300 MPa 26.9 25.7 25.7 23.8 20.2 Tensile strength MPa 27.3 26.3 26.4 26.2 25.7 ε b % 310 310 315 360 425... - Hnbr: 100 g | Source: σ 50 MPa 2.5 2.4 2.3 2.3 2.1 σ 100 MPa 6.1 6.3 5.7 5.7 4.3 σ 200 MPa 18.7*

17.8 17.4 16.1 12.7 σ 300 MPa 26.9 25.7 25.7 23.8 20.2 Tensile strength MPa 27.3 26.3 26.4 26.2 25.7 ε b % 310 310 315 360 425... - 22: 200 MPa | Source: % 62.9 65.4 58.8 63.5 59.0 60.6 62.4 σ 50 MPa 2.4 2.8 2.0 2.6 2.2 2.2 2.3 σ 100 MPa 7.1 8.7 5.3 8.0 5.8 6.5 6.8 σ 200 MPa 22.0 25.9 17.0 24.1 17.7 20.7 22.5 σ 300 MPa — — 26.2 — 26.9 — — Tensile stren... - 24: 2 % | Source: % 62.9 65.4 58.8 63.5 59.0 60.6 62.4 σ 50 MPa 2.4 2.8 2.0 2.6 2.2 2.2 2.3 σ 100 MPa 7.1 8.7 5.3 8.0 5.8 6.5 6.8 σ 200 MPa 22.0 25.9 17.0 24.1 17.7 20.7 22.5 σ 300 MPa — — 26.2 — 26.9 — — Tensile stren... - 12: 200 MPa | Source: % 54.4 57.6 57.6 σ 50 MPa 1.8 2.2 2.3 σ 100 MPa 3.9 5.6 6.2 σ 200 MPa 12.3 17.2 18.3 σ 300 MPa 19.1 24.7 25.2 Tensile strength MPa 24.4 26.4 24.5 E b % 468 337 281 CS (70 h/23° C.); specimen 2 % 20.8 ... - 19: 300 MPa | Source: % 54.4 57.6 57.6 σ 50 MPa 1.8 2.2 2.3 σ 100 MPa 3.9 5.6 6.2 σ 200 MPa 12.3 17.2 18.3 σ 300 MPa 19.1 24.7 25.2 Tensile strength MPa 24.4 26.4 24.5 E b % 468 337 281 CS (70 h/23° C.); specimen 2 % 20.8 ... - 20: 2 % | Source: % 54.4 57.6 57.6 σ 50 MPa 1.8 2.2 2.3 σ 100 MPa 3.9 5.6 6.2 σ 200 MPa 12.3 17.2 18.3 σ 300 MPa 19.1 24.7 25.2 Tensile strength MPa 24.4 26.4 24.5 E b % 468 337 281 CS (70 h/23° C.); specimen 2 % 20.8 ... - 39: 2 % | Source: % 54.4 57.6 57.6 σ 50 MPa 1.8 2.2 2.3 σ 100 MPa 3.9 5.6 6.2 σ 200 MPa 12.3 17.2 18.3 σ 300 MPa 19.1 24.7 25.2 Tensile strength MPa 24.4 26.4 24.5 E b % 468 337 281 CS (70 h/23° C.); specimen 2 % 20.8 ... - 30: 1 % | Source: % 56 53 52 54 54 σ 50 MPa 2.5 2.3 2.4 2.4 2.3 σ 100 MPa 6.1 6.2 6.6 6.8 6.4 σ 200 MPa 18.7 19.4 20 19.9 19.5 Tensile strength MPa 27.3 28.8 29.1 29.1 28.5 ε b % 310 305 300 300 295 CS (70 h/150° C.); ... - 19: 2 % | Source: % 56 53 52 54 54 σ 50 MPa 2.5 2.3 2.4 2.4 2.3 σ 100 MPa 6.1 6.2 6.6 6.8 6.4 σ 200 MPa 18.7 19.4 20 19.9 19.5 Tensile strength MPa 27.3 28.8 29.1 29.1 28.5 ε b % 310 305 300 300 295 CS (70 h/150° C.); ... - Conversion: 15 % | Source: | |---|---|---| Table 4: | | butadiene | 65.5 | |---|---|---| | | acrylonitrile | 33.5/1 | | | Total amount of water | 200 | | | Erkanstol ® BXG1) | 3.67 | | | Baykanol ® PQ2) | 1.10 | | | K s... - Conversion: 36 % | Source: | 0.125 | 0.125 | | *addition at conversion 36%* | | *1)Edenor ® HTiCT N (Oleo Chemicals)* | | *2)trisodium phosphate* 12 H2O (Acros, item number 206520010) - calculation excluding water of crystallizatio... - Emulsifier: 4 kg | Source: | 0.125 | 0.125 | | *addition at conversion 36%* | | *1)Edenor ® HTiCT N (Oleo Chemicals)* | | *2)trisodium phosphate* 12 H2O (Acros, item number 206520010) - calculation excluding water of crystallizatio... Source Text (Relevant Passages): [ABSTRACT] Abstract Novel hydrogenated nitrile rubbers are provided, which, as well as having reduced contents of phosphines and/or diphosphines, include phosphine oxides and a specific content of halogens, and which enable access to vulcanizable mixtures and corresponding vulcanizates that possess improved moduli and compression set values and good values for heat buildup under dynamic stress. These novel hydrogenated nitrile rubbers are obtained by a specific production process, in the course of which the hydrogenated nitrile rubbers are reacted with at least one specific oxidizing agent. [CLAIMS] Claims (20) 1 . Hydrogenated nitrile rubber comprising: i) 0 to 1.0 wt % of phosphines, diphosphines or mixtures thereof, based on the hydrogenated nitrile rubber, ii) 0.25 to 6 wt % of phosphine oxides, diphosphine oxides or mixtures thereof, based on the hydrogenated nitrile rubber, and iii) 25 to 10000 ppm total halogen. 2 . The hydrogenated nitrile rubbers according to claim 1 , wherein; a) the nitrile rubber contains no phosphines or diphosphines; or b) the nitrile rubber comprises greater than 0 to 1.0 wt % of the phosphines, diphosphines, or mixtures thereof, and the nitrile rubbers comprise, as component (i), a phosphine of the general formula (1-a), in which R′ are the same or different and are each alkyl, alkenyl, alkadienyl, alkoxy, aryl, heteroaryl, cycloalkyl, cycloalkenyl, cycloalkadienyl, halogen or trimethylsilyl, and/or a diphosphine of the general formula (1-b), in which R′ are the same or different and are each alkyl, alkenyl, alkadienyl, alkoxy, aryl, heteroaryl, cycloalkyl, cycloalkenyl, cycloalkadienyl, halogen or trimethylsilyl, k is 0 or 1 and X Is a straight-chain or branched alkanediyl, alkenediyl or alkenediyl group. 3 . The hydrogenated nitrile rubbers according to claim 2 , wherein: the phosphine of the general formula (1-a) is PPh_3 , P(p-Tol)$_3$, P(o-Tol)$_3$, $PPh(CH_3)_2$, $P(CF_3)_3$, P(p-FC_6H_4)$_3$, P(p-CF_3CH_4)$_3$, P(C_6H_4—SO_3Na)$_3$, P($CH_2C_6H_4$—SO_3Na)$_3$, P(iso-Pr)$_3$, P(CHCH_3(CH_2CH_3))$_3$, P(cyclopentyl)$_3$, P(cyclo [SYNTHESIS PASSAGES] ted nitrile monomer, at least one conjugated diene and optionally one or more copolymerizable monomers. Processes for producing nitrile rubber and processes for hydrogenating nitrile rubber in suitable organic solvents are known from the literature (e.g. Ullmann's Encyclopedia of Industrial Chemistry, VCH Verlagsgesellschaft, Weinheim, 1993, p. 255-261 and p. 320-324). Hydrogenated nitrile rubber, abbreviated to “HNBR”, is understood to mean rubbers which are obtained from nitrile rubbers, abbreviated to “NBR”, by hydrogenation. Correspondingly, in HNBR, the C■C double bonds of the copolymerized diene units are fully or partly hydrogenated. The hydrogenation level of the ... hemicals and excellent oil resistance. The aforementioned physical and chemical properties of HNBR are combined with very good mechanical properties, especially a high abrasion resistance. Because of this profile of properties, HNBR has found wide use in a wide variety of different areas of application. HNBR is used, for example, for seals, hoses, drive belts, cable sheaths, roller coverings and damping elements in the automotive sector, and also for stators, well seals and valve seals in the oil production sector, and for numerous parts in the aviation industry, the electrical industry, in mechanical engineering and in shipbuilding. A major role is played by vulcanizates ... gations) and low compression set, especially after long storage periods at high temperatures. This combination of properties is important in fields of use in which high resilience forces are required to ensure that the rubber articles will function both under static and under dynamic stress, including after long periods and possibly high temperatures. This applies especially to different seals such as O-rings, flange seals, shaft sealing rings, stators in rotor/stator pumps, valve shaft seals, gasket sleeves such as axle boots, hose seals, engine bearings, bridge bearings and well seals (blowout preventers). In addition, vulcanizates having a high modulus are important, f ... itrile rubber with homogeneously soluble rhodium and/or ruthenium hydrogenation catalysts, the addition of phosphines or diphosphines as a cocatalyst has been found to be useful. Preference is given to the use of triphenylphosphine (“TPP”). This use of a cocatalyst has a number of positive effects: a reduction in the pressure needed for the hydrogenation is enabled; in addition, an increase in the hydrogenation rate (space/time yield) and a reduction in the amount of hydrogenation catalyst needed for the hydrogenation can be achieved. On the other hand, residual amounts of the phosphine or diphosphine remaining in the hydrogenated nitrile rubber have adverse effects on th ... s, especially the modulus level and compression set. DE 25 39 132 A describes a process for hydrogenating random acrylonitrile/butadiene copolymers. For the hydrogenation, the complex of a mono- or trivalent rhodium(I) halide is used in combination with 5 to 25% by weight of triphenylphosphine, with use of 10 phr of triphenylphosphine in each of the examples, DE 25 39 132 A does not vary the amount of triphenylphosphine. Nor are any vulcanizates produced on the basis of the hydrogenated nitrile rubbers or characterized with respect to the properties. DE 25 39 132 A does not give any figures for the contents of triphenylphosphine that remain in the worked-up hydrogenated n ... by sulphur vulcanization is improved by contacting the nitrile rubber after the polymerization or after the hydrogenation with an aqueous alkali solution or the aqueous solution of an amine. In example 1, rubber crumbs that are obtained after removal of the solvent are washed in a separate process step with aqueous sodium carbonate solutions of different concentration. The pH of an aqueous THF solution obtained by dissolving 3 g of the rubber in 100 ml of THF and adding 1 ml of water while stirring is used as a measure of the alkali content. The pH is determined by means of a glass electrode at 20° C. For the production of vulcanizates of the hydrogenated nitrile rubber h ... by removal of TPP after the hydrogenation. U.S. Pat. No. 4,503,196 describes a process for hydrogenating nitrile rubber using rhodium catalysts of the (H)Rh(L) 3 or (H)Rh(L) 4 type. L represents phosphine or arsine ligands. It is a feature of the hydrogenation process that no additions of the ligands as cocatalysts are required for the hydrogenation, although the hydrogenation is effected with relatively high amounts of catalyst (2.5 to 40% by weight). For the isolation of the hydrogenated nitrile rubber from the chlorobenzene solution, the hydrogenated solution is cooled and the rubber is coagulated by addition of isopropanol. U.S. Pat. No. 4,503,196 does not give any i ... vulcanizate properties of the hydrogenated nitrile rubbers that result in this process. For this reason, U.S. Pat. No. 4,503,196 does not give any teaching about the production of hydrogenated nitrile rubber, by which vulcanizates having a high modulus level and low compression set are obtained. DE-A-3 921 264 describes the production of hydrogenated nitrile rubber which, after peroxidic crosslinking, gives vulcanizates having low compression set. For this purpose, ruthenium catalysts of a wide variety of different chemical constitutions are used for the hydrogenation, with use of a solvent mixture of a C 3 -C 6 ketone and a secondary or tertiary C 3 -C 6 alcohol in the ... ed and/or the hydrogenated nitrile rubber gelates during the hydrogenation. The process described in DE-A-3 921 264 is not broadly applicable, since the phase separation which takes place in the course of the hydrogenation and the gelation depends on various parameters in an unpredictable manner. These include the acrylonitrile content and the molar mass of the nitrile rubber feedstock, the composition of the solvent mixture, the solids content of the polymer solution in the hydrogenation, the hydrogenation level and the temperature in the hydrogenation. In the course of cooling of the polymer solution after the hydrogenation or in the course of storage of the polymer sol ... sion set and moduli of vulcanizates can be inferred. EP-A-1 894 946 describes a process for metathesis degradation of NBR, in which the activity of the metathesis catalyst is enhanced by TPP additions. Based on the metathesis catalyst, 0.01 to 1 equivalent of phosphine, for example TPP, is used. Nitrile rubbers of reduced molecular weight prepared in this way can, according to EP-A-1 894 946, be hydrogenated using methods known from the prior art. The catalyst used may, for example, be the Wilkinson catalyst in the presence of cocatalysts such as TPP. There is no information about the vulcanizate properties and the optimization thereof, especially with regard to the level ... uence caused by triphenylphosphine. EP-A-1 524 277 describes an ultrafiltration process for removing low molecular weight constituents from rubbers. For this purpose, the rubbers are dissolved in an organic solvent and subjected to an ultrafiltration process. The process is suitable both for removal of emulsifier residues from nitrile rubber and for removal of catalyst residues from hydrogenated nitrile rubber. According to Example 2 of EP-A-1 524 277, it is possible with the aid of this method to reduce the phosphorus content of hydrogenated nitrile rubber from 1300 mg/kg to 120 mg/kg. EP-A-1 524 277 does not give any information as to whether this process, which is addi ... set of vulcanizates of hydrogenated nitrile rubbers. EP-A-0 134 023 describes a process for hydrogenating NBR with 0.05 to 0.6% by weight, based on rubber solids, of tris(triphenylphosphine)rhodium(I) halide as catalyst, in which not more than 2% by weight, likewise based on rubber solids, of triphenylphosphine is added. The examples in Table 3) show that an increase in the amount of triphenylphosphine to up to 5% by weight leads to a deterioration in important properties of peroxidically vulcanized hydrogenated nitril [EXAMPLES SECTION] example, for seals, hoses, drive belts, cable sheaths, roller coverings and damping elements in the automotive sector, and also for stators, well seals and valve seals in the oil production sector, and for numerous parts in the aviation industry, the electrical industry, in mechanical engineering and in shipbuilding. A major role is played by vulcanizates of hydrogenated nitrile rubber having a high modulus level (measured as the stress value at various elongations) and low compression set, especially after long storage periods at high temperatures. This combination of properties is important in fields of use in which high resilience forces are required to ensure that the rubber articles will function both under static and under dynamic stress, including after long periods and possibly high temperatures. This applies especially to different seals such as O-rings, flange seals, shaft sealing rings, stators in rotor/stator pumps, valve shaft seals, gasket sleeves such as axle boots, hose seals, engine bearings, bridge bearings and well seals (blowout preventers). In addition, vulcanizates having a high modulus are important, for example, for articles under dynamic stress, especially for belts such as drive belts and control belts, for example toothed belts, and also for roller coverings. The level obtained to date in the mechanical properties of HNBR-based vulcanizates, especially in relation to the modulus level and compression set, is still unsatisfactory. For the hydrogenation of nitrile rubber with homogeneously soluble rhodium and/or ruthenium hydrogenation catalysts, the addition of phosphines or diphosphines as a cocatalyst has been found to be useful. Preference is given to the use of triphenylphosphine ("TPP"). This use of a cocatalyst has a number of positive effects: a reduction in the pressure needed for the hydrogenation is enabled; in addition, an increase in the hydrogenation rate (space/time yield) and a reduction in the amount of hydrogenation catalyst needed for the hydrogenation can be achieved. On the other hand, residual amounts of the phosphine or diphosphine remaining in the hydrogenated nitrile rubber have adverse effects on the vulcanizate properties, especially the modulus level and compression set. DE 25 39 132 A describes a process for hydrogenating random acrylonitrile/butadiene copolymers. For the hydrogenation, the complex of a mono- or trivalent rhodium(I) halide is used in combination with 5 to 25% by weight of triphenylphosphine, with use of 10 phr of triphenylphosphine in each of the examples, DE 25 39 132 A does not vary the amount of triphenylphosphine. Nor are any vulcanizates produced on the basis of the hydrogenated nitrile rubbers or characterized with respect to the properties. DE 25 39 132 A does not give any figures for the contents of triphenylphosphine that remain in the worked-up hydrogenated nitrile rubber after the hydrogenation. Nor is there any examination, moreover, of whether and, if so, what influence TPP has on the properties, especially the modulus level and compression set of vulcanizates produced therefrom. Furthermore, there is no indication whatsoever as to the removal of the TPP used in the hydrogenation. In U.S. Pat. No. 4,965,323, the compression set of

HNBR-based vulcanizates which are obtained by peroxidic vulcanization or by sulphur vulcanization is improved by contacting the nitrile rubber after the polymerization or after the hydrogenation with an aqueous alkali solution or the aqueous solution of an amine. In example 1, rubber crumbs that are obtained after removal of the solvent are washed in a separate process step with aqueous sodium carbonate solutions of different concentration. The pH of an aqueous THF solution obtained by dissolving 3 g of the rubber in 100 ml of THF and adding 1 ml of water while stirring is used as a measure of the alkali content. The pH is determined by means of a glass electrode at 20° C. For the production of vulcanizates of the hyd == PATENT: US7919563B2 Title: Low molecular weight hydrogenated nitrile rubber Jurisdiction: US Assignee: Not disclosed Publication Year: 2011-04-05 Source Type: NORMAL Relevance Tier: STRONG (score: 80.0) URL: https://patents.google.com/patent/US7919563B2/en General Parameters: - Concentration: 6 % | Source: Example 4 Cement 15% 15% 6% 6% Concentration Co-olefin C 2 H 4 C 2 H 4 C 2 H 4 1-hexene Co-olefin 500 psi 500 psi 500 psi 150 g Concentration Reactor 80° C - Concentration: 150 g | Source: Example 4 Cement 15% 15% 6% 6% Concentration Co-olefin C 2 H 4 C 2 H 4 C 2 H 4 1-hexene Co-olefin 500 psi 500 psi 500 psi 150 g Concentration Reactor 80° C - Temperature: 0.05 phr | Source: Temperature Catalyst Load 0.05 phr 0.10 phr 0.25 phr 0.25 phr For a typical product the Mn is 27 kg/mol (compared to 85 kg/mol for the starting polymer) while the Mw is 54 kg/mol (compared to 296 kg/m... - Rubber: 200 g | Source: Details 200 g of rubber was dissolved in 1133 g of MCB (15 wt.-% solid) - Mcb: 1133 g | Source: Details 200 g of rubber was dissolved in 1133 g of MCB (15 wt.-% solid) - Rubber: 200 g | Source: Details 200 g of rubber was dissolved in 1133 g of MCB (15 wt.-% solid) - Mcb: 1133 g | Source: Details 200 g of rubber was dissolved in 1133 g of MCB (15 wt.-% solid) - Rubber: 75 g | Source: Details 75 g of rubber was dissolved in 1175 g of MCB (6 wt.-% solid) - Mcb: 1175 g | Source: Details 75 g of rubber was dissolved in 1175 g of MCB (6 wt.-% solid) - Rubber: 75 g | Source: Details 75 g of rubber was dissolved in 1175 g of MCB (6 wt.-% solid) - Mcb: 1175 g | Source: Details 75 g of rubber was dissolved in 1175 g of MCB (6 wt.-% solid) - 1-hexene: 150 g | Source: 150 g of 1-hexene was added to the reactor and the mixture was degassed 3 times with dry N2 under full agitation - 15: 15 % | Source: Example 4 Cement 15% 15% 6% 6% Concentration Co-olefin C 2 H 4 C 2 H 4 C 2 H 4 1-hexene Co-olefin 500 psi 500 psi 500 psi 150 g Concentration Reactor 80° C Examples: [EXAMPLE 3]: - 15: 15 % | Source: Example 4 Cement 15% 15% 6% 6% Concentration Co-olefin C 2 H 4 C 2 H 4 C 2 H 4 1-hexene Co-olefin 500 psi 500 psi 500 psi 150 g Concentration Reactor 80° C - Concentration: 6 % | Source: Example 4 Cement 15% 15% 6% 6% Concentration Co-olefin C 2 H 4 C 2 H 4 C 2 H 4 1-hexene Co-olefin 500 psi 500 psi 500 psi 150 g Concentration Reactor 80° C - Concentration: 150 g | Source: Example 4 Cement 15% 15% 6% 6% Concentration Co-olefin C 2 H 4 C 2 H 4 C 2 H 4 1-hexene Co-olefin 500 psi 500 psi 500 psi 150 g Concentration Reactor 80° C - Temperature: 0.05 phr | Source: Temperature Catalyst Load 0.05 phr 0.10 phr 0.25 phr 0.25 phr For a typical product the Mn is 27 kg/mol (compared to 85 kg/mol for the starting polymer) while the Mw is 54 kg/mol (compared to 296 kg/m... [EXAMPLE 4]: - Rubber: 75 g | Source: Details 75 g of rubber was dissolved in 1175 g of MCB (6 wt.-% solid) - Mcb: 1175 g | Source: Details 75 g of rubber was dissolved in 1175 g of MCB (6 wt.-% solid) - 1-hexene: 150 g | Source: 150 g of 1-hexene was added to the reactor and the mixture was degassed 3 times with dry N2 under full agitation

[EXAMPLE 1]: - Rubber: 200 g | Source: Details 200 g of rubber was dissolved in 1133 g of MCB (15 wt.-% solid) - Mcb: 1133 g | Source: Details 200 g of rubber was dissolved in 1133 g of MCB (15 wt.-% solid) [EXAMPLE 2]: - Rubber: 200 g | Source: Details 200 g of rubber was dissolved in 1133 g of MCB (15 wt.-% solid) - Mcb: 1133 g | Source: Details 200 g of rubber was dissolved in 1133 g of MCB (15 wt.-% solid) [EXAMPLE 3]: - Rubber: 75 g | Source: Details 75 g of rubber was dissolved in 1175 g of MCB (6 wt.-% solid) - Mcb: 1175 g | Source: Details 75 g of rubber was dissolved in 1175 g of MCB (6 wt.-% solid) Source Text (Relevant Passages): [ABSTRACT] Abstract The present invention relates to the process for preparing hydrogenated nitrile rubber polymers having lower molecular weights and narrower molecular weight distributions than those known in the art. [CLAIMS] Claims (16) 1. A process for preparing hydrogenated nitrile rubber having repeating units derived from at least one conjugated diene, at least one α,β-unsaturated nitrile and optionally further one or more copolymerizable monomers, wherein the rubber has a molecular weight (MW) in the range of from 30,000 to 250,000, a Mooney viscosity (ML 1+4 @ 100 deg. C.) in the range of from 3 to 50, and a MWD (or polydispersity index) of less than 2.5 comprising the step of subjecting a nitrile rubber to a methathesis reaction in the presence of one or more compounds of the general formula I, II, III or IV and optionally the presence of a co-olefin and a solvent, wherein, wherein: M is Os or Ru, R and R1 are, independently, hydrogen or a hydrocarbon selected from the group consisting of C 2 -C 20 alkenyl, C 2 -C 20 alkynyl, C 1 -C 20 alkyl, aryl, C 1 -C 20 carboxylate, C 1 -C 20 alkoxy, C 2 -C 20 alkenyloxy, C 2 -C 20 alkynyloxy, aryloxy, C 2 -C 20 alkoxycarbonyl, C 1 -C 20 alkylthio, C 1 -C 20 alkylsulfonyl and C 1 -C 20 alkylsulfinyl, X and X1 are independently any anionic ligand, and L and L1 are independently any neutral ligand or any neutral carbine, optionally, L and L1 can be linked to one another to from a bidentate neutral ligand; wherein M1 is Os or Ru; R2 and R3 are, independently, hydrogen or a hydrocarbon selected from the group consisting of C 2 -C 20 alkenyl, C 2 -C 20 alkynyl, C 1 -C 20 alkyl, aryl, C 1 -C 20 carboxylate, C 1 -C 20 alkoxy, C 2 -C 20 alkenyloxy, C 2 -C 2 [SYNTHESIS PASSAGES] iled Jun. 12, 2001. FIELD OF THE INVENTION The present invention relates to the process of preparing hydrogenated nitrile rubber polymers having lower molecular weights and narrower molecular weight distributions than those known in the art. BACKGROUND OF THE INVENTION Hydrogenated nitrile rubber (HNBR), prepared by the selective hydrogenation of acrylonitrile-butadiene rubber (nitrile rubber; NBR, a co-polymer comprising at least one conjugated diene, at least one unsaturated nitrile and optionally further comonomers), is a specialty rubber which has very good heat resistance, excellent ozone and chemical resistance, and excellent oil resistance. Coupled with the high le ...) content in the range of from 1 to 18% (by IR spectroscopy). One limitation in processing HNBR is the relatively high Mooney Viscosity. In principle, HNBR having a lower molecular weight and lower Mooney viscosity would have better processability. Attempts have been made to reduce the molecular weight of the polymer by mastication (mechanical breakdown) and by chemical means (for example, using strong acid), but such methods have the disadvantages that they result in the introduction of functional groups (such as carboxylic acid and ester groups) into the polymer, and the altering of the microstructure of the polymer. This results in disadvantageous changes in the proper ... microstructure as those rubbers which are currently available, is difficult to manufacture using current technologies. The hydrogenation of NBR to produce HNBR results in an increase in the Mooney viscosity of the raw polymer. This Mooney

Increase Ratio (MIR) is generally around 2, depending upon the polymer grade, hydrogenation level and nature of the feedstock. Furthermore, limitations associated with the production of NBR itself dictate the low viscosity range for the HNBR feedstock. Currently, one of the lowest Mooney viscosity products available is Therban® VP KA 8837 (available from Bayer), which has a Mooney viscosity of 55 (ML 1+4 @ 100° C.) and a RDB of 18%. Kar ... etal salts, in combination with main group alkylating agents, to promote olefin polymerization under mild conditions has had a significant impact on chemical research and production to date. It was discovered early on that some “Ziegler-type” catalysts not only promote the proposed coordination-insertion mechanism but also effect an entirely different chemical process, that is the mutual exchange (or metathesis) reaction of alkenes according to a scheme as shown in FIG. 1 . Acyclic diene metathesis (or ADMET) is catalyzed by a great variety of transition metal complexes as well as non-metallic systems. Heterogeneous catalyst systems based on metal oxides; sulfides or meta ...) and the lack of selectivity resulting from the numerous active sites and side reactions are major drawbacks of the heterogeneous systems. Homogeneous systems have also been devised and used to effect olefin metathesis. These systems offer significant activity and control advantages over the heterogeneous catalyst systems. For example, certain Rhodium based complexes are effective catalysts for the metathesis of electron-rich olefins. The discovery that certain metal-alkylidene complexes are capable of catalyzing the metathesis of olefins triggered the development of a new generation of well-defined, highly active, single-site catalysts. Amongst these, Bis-(tricyclohexyl ... , but the application of olefin metathesis to polymer synthesis has allowed the preparation of new polymeric material with unprecedented properties (such as highly stereoregular poly-norbornadiene). The utilization of olefin metathesis as a means to produce low molecular weight compounds from unsaturated elastomers has received growing interest. The principle for the molecular weight reduction of unsaturated polymers is shown in FIG. 2 . The use of an appropriate catalyst allows the cross-metathesis of the unsaturation of the polymer with the co-olefin. The end result is the cleavage of the polymer chain at the unsaturation sites and the generation of polymer fragments ha ... the lower molecular weight provides good processing behavior. The so-called “depolymerization” of copolymers of 1,3-butadiene with a variety of co-monomers (styrene, propene, divinylbenzene and ethylvinylbenzene, acrylonitrile, vinyltrimethylsilane and divinyldimethylsilane) in the presence of classical Mo and W catalyst system has been investigated. Similarly, the degradation of a nitrile rubber using WCl_6 and $SnMe_4$ or PhC≡CH co-catalyst was reported in 1988. However, the focus of such research was to produce only low molecular fragments, which could be characterized by conventional chemical means and contains no teaching with respect to the preparation of low molecul ... cesses are non-controlled and produce a wide range of products. The catalytic depolymerization of 1,4-polybutadiene in the presence of substituted olefins or ethylene (as chain transfer agents) in the presence of well-defined Grubb's or Schrock's catalysts is also possible. The use of Molybdenum or Tungsten compounds of the general structural formula {M(=NR1)(OR2)2(■CHR); M=Mo, W} to produce low molecular weight polymers or oligomers from gelled polymers containing internal unsaturation along the polymer backbone was claimed in U.S. Pat. No. 5,446,102. Again, however, the process disclosed is non-controlled, and there is no teaching with respect to the preparation of low ... directed to the use of low molecular weight hydrogenated nitrile rubber for the manufacture of a shaped article, such as a seal, hose, bearing pad, stator, well head seal, valve plate, cable sheathing, wheel, roller, pipe seal or footwear component. BRIEF DRAWINGS FIG. 1 illustrates the mutual exchange (or metathesis) reaction of alkenes. FIG. 2 depicts the principle for the molecular weight reduction of unsaturated polymers. DETAILED DESCRIPTION OF THE INVENTION As used throughout this specification, the term “nitrile polymer” is intended to have a broad meaning and is meant to encompass a copolymer having repeating units derived from at least one conj ... iene, at least one α,β-unsaturated nitrile and optionally further one or more copolymerizable monomers. The conjugated diene may be any known conjugated diene, preferably a C 4 -C 6 conjugated diene. Preferred conjugated dienes are butadiene, isoprene, piperylene, 2,3-dimethyl butadiene and mixtures thereof. Even more preferred C 4 -C 6 conjugated dienes are butadiene, isoprene and mixtures thereof. The most preferred C 4 -C 6 conjugated diene is butadiene. The α,β-unsaturated nitrile may be any known α,β-unsaturated nitrile, preferably a C 3 -C 5 α,β-unsaturated nitrile. Preferred C 3 -C 5 α,β-unsaturated nitriles are acrylonitrile, methacrylonitrile, ethacrylonitrile an ... y, the copolymer may further contain repeating units derived from one or more copolymerizable monomers, such as unsaturated carboxylic acids. Non-limiting examples of suitable unsaturated carboxylic acids include fumaric acid, maleic acid, acrylic acid, methacrylic acid and mixtures thereof. Repeating units derived from one or more copolymerizable monomers will replace either the nitrile or the diene portion of the nitrile rubber and it will be apparent to the skilled in the art that the above mentioned weight percents will have to be adjusted to result in 100 weight percent. In case of the mentioned unsaturated carboxylic acids, the nitrile rubber preferably contain repe ... mixtures thereof. The HNBR of the invention is readily available in a two step synthesis, which may take place in the same reaction set-up or different reactors. The metathesis reaction is conducted in the presence of one or more compounds of the general formulas I, II, III or IV; wherein: M is Os or Ru, R and R 1 are, independently, hydrogen or a hydrocarbon selected from the group consisting of C 2 -C 20 alkenyl, C 2 -C 20 alkynyl, C 1 -C 20 alkyl, aryl, C 1 -C 20 carboxylate, C 1 -C 20 alkoxy, C 2 -C 20 alkenyloxy, [EXAMPLES SECTION] example, using strong acid), but such methods have the disadvantages that they result in the introduction of functional groups (such as carboxylic acid and ester groups) into the polymer, and the altering of the microstructure of the polymer. This results in disadvantageous changes in the properties of the polymer. In addition, these types of approaches, by their very nature, produce polymers having a broad molecular weight distribution. A hydrogenated nitrile rubber having a low Mooney (<55) and improved processability, but which has the same microstructure as those rubbers which are currently available, is difficult to manufacture using current technologies. The hydrogenation of NBR to produce HNBR results in an increase in the Mooney viscosity of the raw polymer. This Mooney Increase Ratio (MIR) is generally around 2, depending upon the polymer grade, hydrogenation level and nature of the feedstock. Furthermore, limitations associated with the production of NBR itself dictate the low viscosity range for the HNBR feedstock. Currently, one of the lowest Mooney viscosity products available is Therban® VP KA 8837 (available from Bayer), which has a Mooney viscosity of 55 (ML 1+4 @ 100° C.) and a RDB of 18%. Karl Ziegler's discovery of the high effectiveness of certain metal salts, in combination with main group alkylating agents, to promote olefin polymerization under mild conditions has had a significant impact on chemical research and production to date. It was discovered early on that some “Ziegler-type” catalysts not only promote the proposed coordination-insertion mechanism but also effect an entirely different chemical process, that is the mutual exchange (or metathesis) reaction of alkenes according to a scheme as shown in FIG. 1 . Acyclic diene metathesis (or ADMET) is catalyzed by a great variety of transition metal complexes as well as non-metallic systems. Heterogeneous catalyst systems based on metal oxides; sulfides or metal salts were originally used for the metathesis of olefins. However, the limited stability (especially towards hetero-substituents) and the lack of selectivity resulting from the numerous active sites and side reactions are major drawbacks of the heterogeneous systems. Homogeneous systems have also been devised and used to effect olefin metathesis. These systems offer significant activity and control advantages over the heterogeneous catalyst systems. For example, certain Rhodium based complexes are effective catalysts for the metathesis of electron-rich olefins. The discovery that certain metal-alkylidene complexes are capable of catalyzing the metathesis of olefins triggered the development of a new generation of well-defined, highly active, single-site catalysts. Amongst these, Bis-(tricyclohexylphosphine)-benzylidene ruthenium dichloride (commonly know as Grubb's catalyst) has been widely used, due to its remarkable insensitivity to air and moisture and high tolerance towards various functional groups. Unlike the molybdenum-based metathesis catalysts, this ruthenium carbene catalyst is stable to acids, alcohols, aldehydes and quaternary amine salts and can be used in a variety of solvents (C_6H_6 , CH_2Cl_2 , THF, t-BuOH). The use of transition-metal catalyzed alkene metathesis has since enjoyed increasing attention as a synthetic method. The most commonly used catalysts are based on Mo, W and Ru. Research efforts have been mainly focused on the synthesis of small molecules, but the application of olefin metathesis to polymer synthesis has allowed the preparation of new polymeric material with unprecedented properties (such as highly stereoregular poly-norbornadiene). The utilization of olefin metathesis as a means to produce low molecular weight compounds from unsaturated elastomers has received growing interest. The principle for the molecular weight reduction of unsaturated polymers is shown in FIG. 2 . The use of an appropriate catalyst allows the cross-metathesis of the unsaturation of == PATENT: US9657113B2 Title: Hydrogenation of a diene-based polymer latex Jurisdiction: US Assignee: Not disclosed Publication Year: 2017-05-23 Source Type: NORMAL Relevance Tier: MEDIUM (score: 60.0) URL: https://patents.google.com/patent/US9657113B2/en General Parameters: - Butadiene: 66 % | Source: 8 (100 ml NBR latex, no extra water) b conversions determined by FT-IR; c Defined as mole of hydrogenated C = C units in the NBR per mole of catalyst added per hour Table 1: | TABLE 1 | |---| | Specif... - Solid Content: 19.5 % | Source: (g) (° C.) (MPa) (h) (mol %) (h –1) 1 0.00487 100 6.89 4 98 1810 1a 0.00660 100 6.89 10 22 724 2 0.00244 100 6.89 19 90 760 3 0.00488 120 6.89 1.6 98 4514 4 0.00975 120 6.89 0.5 98 7222 5 0.00125 120... - Solid Content: 19.5 wt % | Source: 8 (100 ml NBR latex, no extra water) b conversions determined by FT-IR; c Defined as mole of hydrogenated C = C units in the NBR per mole of catalyst added per hour Table 1: | TABLE 1 | |---| | Specif... - Pressure: 3.45 MPa | Source: The same procedure as described in example 1 was employed except that a lower hydrogen pressure (500 psi; 3.45 MPa) was used - Pressure: 10.34 MPa | Source: However, a higher hydrogen pressure (1500 psi; 10.34 MPa) was used at 120° C - Parr: 300 mL | Source: The hydrogenation reaction was carried out in a 300 mL Parr 316 Stainless Steel reactor - Wate: 75 ml | Source: First 25 ml of the NBR latex identified in Table 1 and 75 ml water were charged into the reactor - After: 15 % | Source: The hydrogenation degree reached 15% after 4 hours - Resulted: 10 hours | Source: A further extension of the hydrogenation period to 10 hours resulted in a hydrogenation degree of 22% - Hg2: 0.00244 g | Source: The same procedure as described in example 1 was employed except that only 0.00244 g of HG2 catalyst were added - Nbr: 100 ml | Source: 100 ml NBR latex and 0.01253 g of HG2 catalyst were used - Hg2: 0.01253 g | Source: 100 ml NBR latex and 0.01253 g of HG2 catalyst were used - Nbr: 25 ml | Source: (g) (° C.) (MPa) (h) (mol %) (h –1) 1 0.00487 100 6.89 4 98 1810 1a 0.00660 100 6.89 10 22 724 2 0.00244 100 6.89 19 90 760 3 0.00488 120 6.89 1.6 98 4514 4 0.00975 120 6.89 0.5 98 7222 5 0.00125 120... - Nbr: 100 ml | Source: 8 (100 ml NBR latex, no extra water) b conversions determined by FT-IR; c Defined as mole of hydrogenated C = C units in the NBR per mole of catalyst added per hour Table 1: | TABLE 1 | |---| | Specif... Examples: [EXAMPLE]: - Nbr: 25 ml | Source: (g) (° C.) (MPa) (h) (mol %) (h –1) 1 0.00487 100 6.89 4 98 1810 1a 0.00660 100 6.89 10 22 724 2 0.00244 100 6.89 19 90 760 3 0.00488 120 6.89 1.6 98 4514 4 0.00975 120 6.89 0.5 98 7222 5 0.00125 120... - Solid Content: 19.5 % | Source: (g) (° C.) (MPa) (h) (mol %) (h –1) 1 0.00487 100 6.89 4 98 1810 1a 0.00660 100 6.89 10 22 724 2 0.00244 100 6.89 19 90 760 3 0.00488 120 6.89 1.6 98 4514 4 0.00975 120 6.89 0.5 98 7222 5 0.00125 120... - Nbr: 100 ml | Source: 8 (100 ml NBR latex, no extra water) b conversions determined by FT-IR; c Defined as mole of hydrogenated C = C units in the NBR per mole of catalyst added per hour Table 1: | TABLE 1 | |---| | Specif... - Butadiene: 66 % | Source: 8 (100 ml NBR latex, no extra water) b conversions determined by FT-IR; c Defined as mole of hydrogenated C = C units in the NBR per mole of catalyst added per hour Table 1: | TABLE 1 | |---| | Specif... - Solid Content: 19.5 wt % | Source: 8 (100 ml NBR latex, no extra water) b conversions determined by FT-IR; c Defined as mole of hydrogenated C = C units in the NBR per mole of catalyst added per hour Table 1: | TABLE 1 | |---| | Specif... [EXAMPLE 1]: - Parr: 300 mL | Source: The hydrogenation reaction was carried out in a 300 mL Parr 316 Stainless Steel reactor - Wate: 75 ml | Source: First 25 ml of the NBR latex identified in Table 1 and 75 ml water were charged into the reactor - After: 15 % | Source: The hydrogenation degree reached 15% after 4 hours - Resulted: 10 hours | Source: A further extension of the hydrogenation period to 10 hours resulted in a hydrogenation degree of 22% [EXAMPLE 8]: - Nbr: 100 ml | Source: 100 ml NBR latex and 0.01253 g of HG2 catalyst were used - Hg2: 0.01253 g | Source: 100 ml NBR latex and 0.01253 g of HG2 catalyst were used [EXAMPLE 2]: - Hg2: 0.00244 g | Source: The same procedure as described in example 1 was employed except that only 0.00244 g of HG2 catalyst were added [EXAMPLE 6]: - Pressure: 3.45 MPa | Source: The same procedure as described in example 1 was employed except that a lower hydrogen pressure (500 psi; 3.45 MPa) was used [EXAMPLE 7]: - Pressure: 10.34 MPa | Source: However, a higher hydrogen pressure (1500 psi; 10.34 MPa) was used at 120° C [EXAMPLE 3]: (No structured parameters extracted from this example) [EXAMPLE 4]: (No structured parameters extracted from this example) [EXAMPLE 5]: (No structured parameters extracted from this example) Source Text (Relevant Passages): [ABSTRACT] Abstract The present invention provides a novel process for selectively hydrogenating the carbon-carbon double bonds in diene-based polymers which are present in latex form, this means as a suspension of diene-based polymer particles in an aqueous medium, using a Ruthenium or Osmium-based complex catalyst without any organic solvent. [CLAIMS] Claims (10) The invention claimed is: 1. A process for preparing a hydrogenated diene-based polymer, the process comprising subjecting a diene-based polymer in an aqueous suspension to a hydrogenation in the presence of a catalyst of the general formula (A) wherein M is osmium or ruthenium, X 1 and X 2 are identical or different anionic ligands, L is a ligand selected from the group consisting of arsine, stibine, ether, amine, sulphoxide, carboxyl, nitrosyl, pyridine, thioether, and N-heterocyclic carbene ligands, Y is oxygen (O), sulphur (S), an N—R 1 radical or a P—R 1 radical, where R 1 is an alkyl, cycloalkyl, alkenyl, alkynyl, aryl, alkoxy, alkenyloxy, alkynyloxy, aryloxy, alkoxycarbonyl, alkylamino, alkylthio, arylthio, alkylsulphonyl or alkylsulphinyl radical which may in each case optionally be substituted by one or more alkyl, halogen, alkoxy, aryl or heteroaryl radicals, R 2 , R 3 , R 4 and R 5 are identical or different and are each hydrogen or an organic or inorganic radical, and R 6 is hydrogen or an alkyl, alkenyl, alkynyl or aryl radical, wherein the catalyst of the general formula (A) is added in solid form to the aqueous suspension of the diene-based polymer. 2. The process according to claim 1 , wherein the diene-based polymer contains repeating units of at least one (C 4 -C 6) conjugated diene. 3. The process according to claim 2 , wherein the diene-based polymer additionally contains repeating units of at least one further copolymerizable monomer (b). 4 [SYNTHESIS PASSAGES] Description FIELD OF THE INVENTION The present invention relates to a process for selectively hydrogenating the carbon-carbon double bonds in diene-based polymers which are present in latex form, this means as a suspension of diene-based polymer particles in an aqueous medium, using a ruthenium- or osmium-based metathesis catalyst without any organic solvent. BACKGROUND OF THE INVENTION It has been known that carbon-carbon double bonds in polymers may be successfully hydrogenated by treating the polymer in an organic solution with hydrogen in the presence of different catalysts. Such processes can be selective in the double bonds which are hydrogenated so that, ... nitrogen or oxygen are not affected. This field of art contains many examples of catalysts suitable for such hydrogenations e.g. based on cobalt, nickel, rhodium, ruthenium, osmium, and iridium. The suitability of the catalyst depends on the extent of hydrogenation required, the rate of the hydrogenation reaction and the presence or absence of other groups, such as carboxyl and nitrile groups, in the polymers. Hydrogenation of diene-based polymers has been very successful, if the polymers were dissolved in an organic solvent as e.g. disclosed in U.S. Pat. No. 6,410,657, U.S. Pat. No. 6,020,439, U.S. Pat. No. 5,705,571, U.S. Pat. No. 5,057,581, and U.S. Pat. No. 3,454,644. ... er, many diene-based polymers, -copolymers or -terpolymers are made by emulsion polymerization processes and are therefore obtained in latex form, i.a. as polymer particles suspended in the aqueous medium due to the stabilizing effect of emulsifiers, when they are discharged from polymerization reactors. Therefore it is very desirable to directly hydrogenate a diene-based polymer in said latex form and increasing efforts are spent on such direct hydrogenation in the recent decade. So far significant attention has been paid to the hydrogenation of C■C bonds using hydrazine or a derivative of hydrazine as a reducing agent together with an oxidant like oxygen, air or hydroge ... ,039,737 and U.S. Pat. No. 5,442,009 provide a more refined latex hydrogenation process which treats the hydrogenated latex with ozone to break the cross-linked polymer chains which form during or after the latex hydrogenation using the diimide approach. U.S. Pat. No. 6,552,132 discloses that specific compounds, if added before, during or after the latex hydrogenation serve to break cross-links formed during the hydrogenation using the diimide hydrogenation route. The specific compound can be chosen from primary or secondary amines, hydroxylamine, imines, azines, hydrazones and oximes. U.S. Pat. No. 6,635,718 describes the process for hydrogenating C■C bonds of an unsatur ... n efficiency and degree of hydrogenation. It has been found that there are side reactions at the inter phase of the latex particles and within the polymer phase, which generate radicals to initiate the cross-linking of polymers in the latex form. Using radical scavengers did not show any evidence in helping to suppress the degree of gel formation. Although there are methods developed to reduce the cross-linking, the aforementioned diimide route still encounters gel formation problem, especially when high hydrogenation conversion is achieved. Therefore, the resulting hydrogenated rubber mass is difficult to process or is unsuitable for further use because of its macroscopi ... a hydrogenation catalyst being a palladium compound. In this process an aqueous emulsion of the unsaturated, nitrile-group-containing polymer is subjected to the hydrogenation and additionally an organic solvent capable of dissolving or swelling the polymer is used at a volume ratio of the aqueous emulsion to the organic solvent in a range of from 1:1 to 1:0.05. The aqueous emulsion is brought into contact with gaseous or dissolved hydrogen while maintaining an emulsified state. U.S. Pat. No. 6,403,727 discloses a process for selectively hydrogenating ethylenically unsaturated double bonds in polymers. Said process involves reacting the polymers with hydrogen in the prese ... ises up to 20% by volume of an organic solvent. The suitable rhodium containing catalysts are rhodium phosphine complexes of the formula RhX m L 3 L 4 (L 5) n wherein X is a halide, the anion of a carboxylic acid, acetylacetonate, aryl- or alkylsulfonate, hydride or the diphenyltriazine anion and L 3 , L 4 and L 5 independently are CO, olefins, cycloolefins, dibenzophosphol, benzonitrile, PR 3 or R 2 P-A-PR 2 , m is 1 or 2 and n is 0, 1 or 2, with the proviso that at least one of L 3 , L 4 or L 5 is one of the above mentioned phosphorus-containing ligands of the formula PR 3 or PR 2 -A-PR 2 , wherein R is alkyl, alkyloxy, cycloalkyl, cycloalkyloxy, aryl or aryloxy. U.S. ... catalyst in order to prepare graft polymers. JP 2001-288212 describes a further process for hydrogenating diene-based polymer latices. Latices of 2-chloro-1,3-butadiene (co)polymers are mixed with solutions or dispersions of catalysts in organic solvents which dissolve or swell the (co)polymers, and then contacted with hydrogen. The catalysts used are the so-called Wilkinson-catalysts having the formula MeCl a (P(C 6 H 5) 3) b wherein Me is a transition-metal, Cl is chlorine, b is an integer and equal to or bigger than 1 and a+b is an integer less than or equal to 6. In the Examples a latex of poly(2-chloro-1,3-butadiene) rubber having a T g of –42° C. and an average nu ... he selective hydrogenation of the C■C bonds in nitrile-butadiene rubber (“NBR”) emulsions are described, which are carried out in the presence of a number of RuCl 2 (PPh 3) 3 complex catalysts. One of the processes is carried out in a homogeneous system, in which an organic solvent, which can dissolve the NBR polymer and the catalyst and which is compatible with the emulsion, is used. The other process is carried out in a heterogeneous system, in which an organic solvent, which is capable of dissolving the catalyst and swelling the polymer particles but is not miscible with the aqueous emulsion phase, is used. Both processes can realize quantitative hydrogenation of the ... ic solvent to dissolve or swell the polymers. U.S. Pat. No. 6,696,518 teaches a process for selective hydrogenation of non-aromatic C■C and C■C bonds in polymers with hydrogen in the presence of at least one hydrogenation catalyst comprising ruthenium and/or rhodium and at least one nonionic phosphorus compound capable of forming a coordinative compound with the transition metal wherein the hydrogenation catalyst is incorporated into the aqueous dispersion of the polymer without adding a solvent. Ru and/or Ru complexes or Ru and/or Ru salts are used as catalysts. Examples of preferred nonionic phosphorus compound are PR 3 or R 2 P(O) x Z(O) y PR 2 [R represents e.g. C 1-1 ... case, an acrylic acid-butadiene-styrene copolymer latex was prepared by radical polymerization of a mixture of monomers containing also ruthenium(III) tris-2,4-pentanedionate, which means the Ru salt was dispersed into monomer aqueous solution as the catalyst precursor before the polymerization. After having obtained the aqueous polymer dispersion, Bu 3 P was added to the latex. The system was stirred for 16 h at ambient temperature followed by hydrogenation at severe conditions for 30 hours at 150° C. and 280 bar. The catalyst was thereby synthesized in-situ, therefore no organic solvent was used to transport the catalyst. The hydrogenation is carried out in aqueous disp ... 6,696,518, i.e. adding the catalyst precursor to the monomer mixture before the polymerization takes place, is associated with some problems, including that the catalyst precursor may have a negative effect on the polymerization and that some of the catalyst precursor might get deactivated during the polymerization. In J. Molecular Catalysis Vol. 123, no. 1, 1997, 15-20 it is reported on the hydrogenation of polybutadiene (PBD), as well as polymers having styrene-butadiene repeating units (SBR) or having nitrile-butadiene repeat [EXAMPLES SECTION] example, the double bonds in aromatic or naphthenic groups are not hydrogenated and double or triple bonds between carbon and other atoms such as nitrogen or oxygen are not affected. This field of art contains many examples of catalysts suitable for such hydrogenations e.g. based on cobalt, nickel, rhodium, ruthenium, osmium, and iridium. The suitability of the catalyst depends on the extent of hydrogenation required, the rate of the hydrogenation reaction and the presence or absence of other groups, such as carboxyl and nitrile groups, in the polymers. Hydrogenation of diene-based polymers has been very successful, if the polymers were dissolved in an organic solvent as e.g. disclosed in U.S. Pat. No. 6,410,657, U.S. Pat. No. 6,020,439, U.S. Pat. No. 5,705,571, U.S. Pat. No. 5,057,581, and U.S. Pat. No. 3,454,644. However, many diene-based polymers, -copolymers or -terpolymers are made by emulsion polymerization processes and are therefore obtained in latex form, i.a. as polymer particles suspended in the aqueous medium due to the stabilizing effect of emulsifiers, when they are discharged from polymerization reactors. Therefore it is very desirable to directly hydrogenate a diene-based polymer in said latex form and increasing efforts are spent on such direct hydrogenation in the recent decade. So far significant attention has been paid to the hydrogenation of C■C bonds using hydrazine or a derivative of hydrazine as a reducing agent together with an oxidant like oxygen, air or hydrogen peroxide. The hydrogen source to saturate the C■C bonds is then generated in-situ as a result of the redox reactions in which diimide is also formed as intermediate. In U.S. Pat. No. 4,452,950 the latex hydrogenation is performed using the hydrazine hydrate/hydrogen peroxide (or oxygen) redox system to produce diimide in situ. CuSO 4 or FeSO 4 is used as a catalyst. U.S. Pat. No. 5,039,737 and U.S. Pat. No. 5,442,009 provide a more refined latex hydrogenation process which treats the hydrogenated latex with ozone to break the cross-linked polymer chains which form during or after the latex hydrogenation using the diimide approach. U.S. Pat. No. 6,552,132 discloses that specific compounds, if added before, during or after the latex hydrogenation serve to break cross-links formed during the hydrogenation using the diimide hydrogenation route. The specific compound can be chosen from primary or secondary amines, hydroxylamine, imines, azines, hydrazones and oximes. U.S. Pat. No. 6,635,718 describes the process for hydrogenating C■C bonds of an unsaturated polymer in the form of

an aqueous dispersion by using hydrazine and an oxidizing compound in the presence of a metal compound containing a metal atom in an oxidation state of at least 4 (such as Ti(IV), V(V), Mo(VI) and W(VI)) as the catalyst. In Applied Catalysis A-General Vol. 276, no. 1-2, 2004, 123-128 and Journal of Applied Polymer Science Vol. 96, no. 4, 2005, 1122-1125 detailed investigations relating to the hydrogenation of nitrile butadiene rubber latex via the diimide hydrogenation route are presented which cover examining hydrogenation efficiency and degree of hydrogenation. It has been found that there are side reactions at the inter phase of the latex particles and within the polymer phase, which generate radicals to initiate the cross-linking of polymers in the latex form. Using radical scavengers did not show any evidence in helping to suppress the degree of gel formation. Although there are methods developed to reduce the cross-linking, the aforementioned diimide route still encounters gel formation problem, especially when high hydrogenation conversion is achieved. Therefore, the resulting hydrogenated rubber mass is difficult to process or is unsuitable for further use because of its macroscopic three dimensional cross-linked structure. U.S. Pat. No. 5,272,202 describes a process for the selective hydrogenation of the carbon-carbon double bonds of an unsaturated, nitrile-group
== PATENT: EP4221260B1 Title: MEMBRAN AND SOUND GENERATION DEVICE Jurisdiction: EP Assignee: Not disclosed Publication Year: 2025-12-31 Source Type: NORMAL Relevance Tier: MEDIUM (score: 30.0) URL: https://patents.google.com/patent/EP4221260B1/en General Parameters: - Temperature: 80 °C | Source: In an embodiment, the diaphragm has an operating temperature in the range of -80°C to 180°C - Time: 90 % | Source: Effect of the content of the vulcanizing agent on the glass transition temperature and elongation at break of the material Content of the Vulcanizing Agent (wt%) 0.1 0.5 1 5 15 Glass transition temper... - Tensile: 100 % | Source: The relationship between carbon black particle size and the specific surface area, tear strength, 100% tensile modulus and product F0 Examples: [GENERAL PROCEDURE 1]: - Temperature: 80 °C | Source: In an embodiment, the diaphragm has an operating temperature in the range of -80°C to 180°C - Tensile: 100 % | Source: The relationship between carbon black particle size and the specific surface area, tear strength, 100% tensile modulus and product F0 - Time: 90 % | Source: Effect of the content of the vulcanizing agent on the glass transition temperature and elongation at break of the material Content of the Vulcanizing Agent (wt%) 0.1 0.5 1 5 15 Glass transition temper... Source Text (Relevant Passages): [CLAIMS] Claims (13) A diaphragm comprising at least one elastomer layer, the elastomer layer is made of a rubber compound, the rubber compound comprises nitrile rubber and hydrogenated nitrile rubber, both the nitrile rubber and the hydrogenated nitrile rubber contain acrylonitrile blocks, and in terms of mass percentage, the range of the content of the acrylonitrile blocks is 20% to 80%. The diaphragm according to claim 1, wherein a mass ratio of the nitrile rubber and the hydrogenated nitrile rubber is no less than 1. The diaphragm according to claim 1, wherein the rubber compound further comprises filler, and in terms of mass percentage, the content of the filler in the rubber compound is 5% to 60%. The diaphragm according to claim 3, wherein the filler has a particle size in the range of 10nm to 10μm; and/or, the filler is at least one selected from the group consisting of carbon black, white carbon black, calcium carbonate, barium sulfate, montmorillonite, and mica. The diaphragm according to claim 1, wherein the rubber compound further comprises a vulcanizing agent, and

in terms of mass percentage, the content of the vulcanizing agent in the rubber compound is 0.5% to 5%. The diaphragm according to claim 5, wherein the vulcanizing agent comprises a main cross-linking agent and a co-cross-linking agent, the main cross-linking agent has a vulcanizing system of at least one of sulfur vulcanizing systems and peroxide vulcanizing systems, and the co-cross-linking agent is at least [SYNTHESIS PASSAGES] tic resin has poor heat and cold resistance, plastic deformation occurs at high temperature, and it is brittle at low temperature. SUMMARY The present invention is directed to a diaphragm defined according to independent claim 1. Further aspects of the invention are defined according to the dependent claims. The main object according to the present invention is to provide a diaphragm and a sound generating apparatus, it aims to improve the comprehensive performance of the diaphragm, so as to have excellent mechanical strength, cold resistance, resilience and high damping performance. To achieve the above object, the diaphragm provided in the present invention comprises at ... he range of 15N/mm to 100N/mm. In an embodiment, the diaphragm has an operating temperature in the range of -80°C to 180°C. In an embodiment, the damping factor of the diaphragm is greater than 0.15. In an embodiment, the diaphragm has a density in the range of 0.9g/cm 3 to 1.2g/cm 3 . In an embodiment, the diaphragm is a single layer, which is made of the rubber compound. Alternatively, the diaphragm is a composite layer, and at least one of the composite layer is made of the rubber compound. The present invention also provides a sound generating apparatus which comprises a magnetic circuit system and a vibration system. The vibration system comprises the diaphragm, and ... on, the diaphragm can be used normally under extreme conditions of high and low temperatures, a phenomenon of film rupture is not easy to occur, and the service life of the diaphragm is longer than the conventional diaphragm material, and the diaphragm has better reliability. The sound generating apparatus applying the diaphragm can be applied in extremely harsh circumstance, while maintaining good acoustic performance. Moreover, the content of the acrylonitrile blocks is reasonably adjusted so that the mass content of the acrylonitrile blocks ranges from 20% to 80%, and such prepared diaphragm has excellent cold resistance, mechanical strength and resilience at the same ... iaphragm according to the embodiment of present invention before and after high temperatures; Fig. 2 is a schematic diagram of the comparison curve of resonant frequency F0 of a conventional diaphragm before and after high temperatures; Fig. 3 is a total harmonic distortion test curve of the loudspeaker diaphragm according to an embodiment of the present invention and a conventional diaphragm. The realization of the object, functional features and advantages of the present invention will be further described with reference to the attached drawings in combined with embodiments. DETAILED DESCRIPTIONS The present invention provides a diaphragm which is applied to a sound gen ... e content of the acrylonitrile blocks ranges from 20% to 80%. Nitrile rubber is polymerized from acrylonitrile and butadiene, its molecular formula comprises acrylonitrile blocks and butadiene blocks. The butadiene blocks provide flexibility in the material substrate, so that the nitrile rubber has good low temperature resistance properties. While the nitrile group of the acrylonitrile blocks is a strong polar group, which has high electronegativity, and it can form hydrogen bonds with atoms in the vulcanizing agent, so that the activity of the molecular chain is limited. In addition, the greater the content of the acrylonitrile, the more such alternating structural units ... Hydrogenated nitrile rubber is a highly saturated special elastomer obtained by hydrogenation of nitrile rubber, which has excellent thermo-oxidative aging resistance properties. The molecular structure of hydrogenated nitrile rubber also comprises acrylonitrile blocks. It is understand that, the nitrile group is a strong polar group having high electronegativity, and as the content of acrylonitrile blocks increases, the polarity of the nitrile group is enhanced, the chain flexibility is reduced, and the inter-chain interaction force is increased. The higher the glass transition temperature is, the content of double bond within the molecular chain decreases, the degree of ... of vulcanization is increased, and the tensile strength is increased, while the processability and cold resistance are decreased, and the resilience is decreased. A prepared diaphragm with poor cold resistance is easy to become stiff and brittle in an extreme circumstance of low temperatures, and the risk of diaphragm rupture increases. Therefore, the content of acrylonitrile blocks should be controlled properly, so that the prepared diaphragm has excellent glass transition temperature, tensile strength, density and resilience. Referring to Table 1, Table 1 shows the effect of the content of acrylonitrile blocks on the diaphragm performance. It can be seen from Table 1, a ... ilience decreases sharply. In order to ensure it has excellent glass transition temperature, tensile strength, density and resilience, in terms of mass percentage, the content of acrylonitrile blocks ranges from 20% to 80%. For example, the range of the content of acrylonitrile blocks is 20%, 30%, 40%, 50%, 60%, 70%, or 80%. Table 1. Effect of the content of the acrylonitrile blocks on diaphragm performance The content of Acrylonitrile Blocks (wt%) 5 20 50 80 85 Glass Transition Temperature (°C) -35.1 -31.1 -23.5 -17.1 -15.6 Tensile Strength (MPa) 22.5 28.3 32.9 35 40.4 Density (g/cm 3) 0.91 0.96 1.11 1.19 1.35 Resilience (%) 88 85 81 78 62 Therefore, it is understand th ... e at the same time. Due to the high price of hydrogenated nitrile rubber in the preparation of the rubber compound, in an optional embodiment, the mass ratio of nitrile rubber and hydrogenated nitrile rubber is not less than 1, so that the cost of the rubber compound may be reduced to a certain extent, thereby reducing the production cost of the diaphragm. Further, the rubber compound further comprises filler. In terms of mass percentage, the content of the filler in the rubber compound is 5% to 60%. Generally, the filler is inorganic filler with hydrogen, carboxyl group, lactone group, radical group, quinonyl group, and the like that can subject to substitution, reductio ... and the like, which can subject to substitution, reduction, oxidation and other reactions. When carbon black is added into the rubber compound, since the strong interaction between the surface of carbon black and the interface of the rubber compound, when the material is stressed, the molecular chain is easier to slide on the surface of carbon black, while it is not easy to detach from carbon black. The rubber compound and carbon black form a strong bond that can slide, and the mechanical strength increases. It should be noted that when a filler fortifier is added, the content of the filler fortifier added needs to be reasonably controlled, such that the damping is approp ... suppressed, the sound quality is good and the distortion is low, the mechanical properties are superior, and the acoustic performance and reliability conditions are satisfied. Optionally, in terms of mass percentage, the content of filler fortifier in rubber compound ranges from 5% to 60%, for example, the content of filler fortifier in rubber compound is 5%, 10%, 20%, 25%, 35%, 45%, 55%, or 60%. Referring to Table 2, Table 2 shows the effect of the amount of the carbon black added on the mechanical properties of the material. It can be seen from the table, when the parts of carbon black is 0.5%, both the mechanical strength and elongation at break are small, it is becaus ... ation system that can suppress the polarization phenomenon during the vibration process is strong, and the vibration consistency is good. The commonly used engineering plastic film layer has low damping, and its damping factor is generally less than 0.01, and the damping is small. In the case of an excessive amount of the carbon black filled, the hysteresis of the rubber is greater than the optimal added amount in the rubber, and the reinforcement effect is reduced, and high damping performance will make the heat gen [EXAMPLES SECTION] example, the high modulus plastic film layer is difficult to withstand large mechanical vibration in high frequency circumstance, thereby a phenomenon of film rupture occurs; the thermoplastic resin has poor heat and cold resistance, plastic deformation occurs at high temperature, and it is brittle at low temperature. SUMMARY The present invention is directed to a diaphragm defined according to independent claim 1. Further aspects of the invention are defined according to the dependent claims. The main object according to the present invention is to provide a diaphragm and a sound generating apparatus, it aims to improve the comprehensive performance of the diaphragm, so as to have excellent mechanical strength, cold resistance, resilience and high damping performance. To achieve the above object, the diaphragm provided in the present invention comprises at least one elastomer layer, and the elastomer layer is made of a rubber compound. The rubber compound comprises nitrile rubber and hydrogenated nitrile rubber, and both the nitrile rubber and the hydrogenated nitrile rubber contain acrylonitrile blocks. In terms of mass percentage, the range of the content of acrylonitrile blocks is 20% to 80%. In an embodiment, the mass ratio of the nitrile rubber and the hydrogenated nitrile rubber is no less than 1. In an embodiment, the rubber compound further comprises filler. In terms of mass percentage, the content of the filler in the rubber compound is 5% to 60%. In an embodiment, the filler has a particle size in the range of 10nm to10μm; and/or, the filler is at least one selected from the group consisting of carbon black, white carbon black, calcium carbonate, barium sulfate, montmorillonite, and mica. In an embodiment, the rubber compound further comprises a vulcanizing agent. In terms of mass percentage, the content of the vulcanizing agent in the rubber compound is 0.5% to 5%. In an embodiment, the vulcanizing agent comprises a main cross-linking agent and a co-cross-linking agent. The vulcanizing system of the main cross-linking agent is at least one of sulfur vulcanizing systems and peroxide vulcanizing systems, and the co-cross-linking agent is at least one of triallyl isocyanurate and polymaleimide. In an embodiment, the diaphragm has a tensile strength in the range of 4MPa to 25MPa. In an embodiment, the diaphragm has a tear strength in the range of 15N/mm to 100N/mm. In an embodiment, the diaphragm has an operating temperature in the range of -80°C to 180°C. In an embodiment, the damping factor of the diaphragm is greater than 0.15. In an embodiment, the diaphragm has a density in the range of 0.9g/cm 3 to 1.2g/cm 3 . In an embodiment, the diaphragm is a single layer, which is made of the rubber compound. Alternatively, the diaphragm is a composite layer, and at least one of the composite layer is made of the rubber compound. The present invention also provides a sound generating apparatus which comprises a magnetic circuit system and a vibration system. The vibration system comprises the diaphragm, and the diaphragm comprises at least one elastomer layer. The elastomer layer is made of a rubber compound, and the rubber compound comprises nitrile rubber and hydrogenated nitrile rubber. Both the nitrile rubber and the hydrogenated nitrile rubber contain acrylonitrile blocks, and in terms of mass percentage, the range of the content of the acrylonitrile blocks is 20% to 80%. The diaphragm according to the technical solution of the present invention adopts at least one elastomer layer, the elastomer layer is made of a rubber compound, and the rubber compound comprises nitrile rubber and hydrogenated nitrile rubber. Therefore, the diaphragm may exhibit a better comprehensive performance, and has excellent mechanical strength, cold resistance, resilience and high damping performance. In addition, the diaphragm can be used normally under extreme conditions of high and low temperatures, a phenomenon of film rupture is not easy == PATENT: EP3030824B1 Title: Method for the production of and high temperature field joints Jurisdiction: EP Assignee: Not disclosed Publication Year: 2024-03-27 Source Type: NORMAL Relevance Tier: MEDIUM (score: 30.0) URL: https://patents.google.com/patent/EP3030824B1/en General Parameters: None extracted by deterministic parser. Examples: [EXAMPLE]: (No structured parameters extracted from this example) Source Text (Relevant Passages): [CLAIMS] Claims (16) A method for forming a field joint between two insulated pipe sections (10, 12), the method comprising: (a) providing a first insulated pipe section (10) and a second insulated pipe section (12), wherein each of the insulated pipe sections (10, 12) comprises: (i) a steel pipe (14) having an outer surface (16) and an end (15), wherein an annular connection surface (18) is located at said end (15) of the steel pipe (14), (ii) a corrosion protection coating (20) provided over the outer surface (16) of the steel pipe (14), wherein a terminal end of the corrosion protection coating (20) is spaced from the end (15) of the pipe (14); and (iii) a pipe insulation layer (22) provided over the corrosion protection coating (20), wherein a terminal end (26) of the pipe insulation layer (22) is spaced from the end (15) of the pipe (14), and wherein the pipe insulation layer (22) comprises a polymer composition having thermal conductivity of less than about 0.40 W/mk, and/or heat resistance to continuous operating temperatures from about 150°C to above about 205°C; wherein the corrosion protection coating (20) and the pipe insulation layer (22) together comprise a line pipe coating of the insulated pipe sections (10, 12); and wherein each of the insulated pipe sections (10, 12) has a bare end portion (24) in which the outer surface (16) of the steel pipe (14) is exposed, the bare end portion (24) extending from the end (15) of the steel pipe (14) to the terminal end of either [SYNTHESIS PASSAGES] having factory-applied corrosion protection and insulating coatings. During the manufacture of insulated pipe, the ends of the pipe must be left bare to prevent damage to the coating when the pipes are joined in the field by welding. Typically, the insulation layer is cut back from the end of the pipe to form a chamfer which is spaced from the end of the pipe. A lip of the corrosion protection layer may protrude beyond the end (or

Polymerization / Synthesis Method - Monomers: Conjugated diene (40% to 90% by weight), \u03b1,\u03b2-unsaturated nitrile (10% to 60% by weight) - Catalyst: Ruthenium compound (10 ppm to 200 ppm, preferably 42 ppm to 79 ppm), Palladium compound (20 ppm to <67 ppm), or Rhodium compound (50 ppm to <270 ppm) - Temperature: 60 to 200\u00b0 C. - Pressure: 700,000 to 15,000,000 pascals - Time: 1 to 24 hours

Relevant Experimental Evidence - The patent describes a method where NBR in solution is hydrogenated using low amounts of noble metal catalysts (e.g., 10-200 ppm Ru, 20-<67 ppm Pd, or 50-<270 ppm Rh) at 60-200\u00b0C and 0.7-15 MPa for 1-24 hours. - The resulting HNBR exhibits improved ageing properties, specifically characterized by a gel content increase of less than 10% and a Mooney viscosity increase of less than 40% after ageing for 4 days at

140\u00b0C. - No specific experimental examples with numbered runs are detailed in the provided text, but the general parameters for the hydrogenation of NBR containing 40-90 wt% conjugated diene and 10-60 wt% unsaturated nitrile are defined.

Technical Relevance Provides a method to synthesize HNBR with excellent ageing properties by controlling the residual catalyst concentration (Ru, Pd, or Rh) to very low levels, eliminating the need for complex catalyst removal steps.

Patent 2

- Patent Number: US20160326272A1
- Patent Title: Hydrogenated nitrile rubber containing phosphine oxide or diphosphine oxide
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2016
- Relevance: Describes HNBR containing phosphine oxides/diphosphine oxides and halogens to improve modulus, compression set, and heat buildup.

Polymerization / Synthesis Method - Monomers: Butadiene (65.5 parts by weight), Acrylonitrile (33.5 parts by weight) - Water: 200 parts by weight - Emulsifier: Erkanstol\u00ae BXG (3.67 parts by weight), Baykanol\u00ae PQ (1.10 parts by weight) - Chain Transfer Agent: TDM (0.125 parts by weight added at 36% conversion) - Co-catalyst: Triphenylphosphine (2.9 parts by weight per 100 g NBR) - Pressure: 120 bar - Temperature: Roll temperature of 20\u00b0 C.

Relevant Experimental Evidence - Examples describe a fully hydrogenated nitrile rubber with 39% by weight of acrylonitrile, hydrogenated using 2.9 parts by weight of triphenylphosphine (TPP) per 100 g of NBR. - The HNBR is reacted with oxidizing agents (such as potassium chlorate, potassium perchlorate, potassium permanganate, potassium dichromate, or 1,4-benzoquinone) in the solid state to convert residual phosphines to phosphine oxides. - The resulting HNBR comprises 0 to 1.0 wt% phosphines/diphosphines, 0.25 to 6 wt% phosphine oxides/diphosphine oxides, and 25 to 10,000 ppm total halogen. - Vulcanizates prepared from these treated rubbers show improved moduli (e.g., stress values at 50%, 100%, 200%, and 300% strain), lower compression set (e.g., after 70 h at 150\u00b0C), and reduced heat buildup.

Technical Relevance Discloses a post-treatment method for HNBR to oxidize residual phosphine cocatalysts (used during homogeneous hydrogenation) into phosphine oxides, which significantly improves the mechanical and dynamic properties of the vulcanizates.

Patent 3

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US

- Publication Year: 2011
- Relevance: Describes a process for preparing low molecular weight HNBR with narrow molecular weight distribution via metathesis followed by hydrogenation.

Polymerization / Synthesis Method - Starting Polymer: Nitrile rubber (NBR) - Solvent: Monochlorobenzene (MCB) - Concentration: 6 wt% or 15 wt% solid - Co-olefin: Ethylene (C2H4) or 1-hexene - Co-olefin Pressure: 500 psi - Co-olefin Amount: 150 g (for 1-hexene) - Catalyst Load: 0.05 phr, 0.10 phr, or 0.25 phr - Reactor Temperature: 80\u00b0 C.

Relevant Experimental Evidence - Example 1 & 2: 200 g of NBR was dissolved in 1133 g of monochlorobenzene (MCB) to form a 15 wt% solid solution. - Example 3 & 4: 75 g of NBR was dissolved in 1175 g of MCB to form a 6 wt% solid solution. - In Example 4, 150 g of 1-hexene was added as a co-olefin, and the mixture was degassed 3 times with dry N2. - Metathesis was conducted using a ruthenium-based Grubbs catalyst (loads of 0.05 to 0.25 phr) at 80\u00b0C under co-olefin pressure (e.g., 500 psi ethylene). - The process successfully reduced the molecular weight of the polymer (e.g., Mn from 85 kg/mol to 27 kg/mol, Mw from 296 kg/mol to 54 kg/mol) and yielded a polydispersity index of less than 2.5 and a Mooney viscosity (ML 1+4 @ 100\u00b0C) of 3 to 50.

Technical Relevance Demonstrates a two-step synthesis (metathesis degradation followed by hydrogenation) to produce low-Mooney HNBR with excellent processability without altering the polymer's microstructure or introducing unwanted functional groups.

Patent 4

- Patent Number: US9657113B2
- Patent Title: Hydrogenation of a diene-based polymer latex
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2017
- Relevance: Describes a process for selectively hydrogenating diene-based polymers (like NBR) in latex form using a ruthenium or osmium-based catalyst without organic solvents.

Polymerization / Synthesis Method - Starting Material: NBR latex (solid content 19.5 wt%, butadiene content 66%) - Latex Volume: 25 ml or 100 ml - Water: 75 ml (added to 25 ml latex in Example 1) - Catalyst: HG2 (Hoveyda-Grubbs 2nd generation) catalyst (0.00125 g to 0.01253 g) - Temperature: 100\u00b0 C. or 120\u00b0 C. - Pressure: 3.45 MPa (500 psi), 6.89 MPa (1000 psi), or 10.34 MPa (1500 psi) - Time: 0.5 to 19 hours - Conversion: 22% to 98% hydrogenation degree

Relevant Experimental Evidence - Example 1: 25 ml of NBR latex (19.5 wt% solid content) and 75 ml water were charged into a 300 mL Parr reactor. 0.00487 g of HG2 catalyst was added in solid form. Hydrogenation at 100\u00b0C and 6.89 MPa for 4 hours yielded a 98% hydrogenation degree. - Example 1a: Using 0.00660 g of catalyst at 100\u00b0C and 6.89 MPa for 10 hours resulted in only 22% hydrogenation degree (showing sensitivity to conditions). - Example 2: Using

0.00244 g of HG2 catalyst at 100\u00b0C and 6.89 MPa for 19 hours achieved a 90% hydrogenation degree. - Example 3 & 4: Conducted at 120\u00b0C and 6.89 MPa. Example 3 (0.00488 g catalyst) reached 98% conversion in 1.6 hours. Example 4 (0.00975 g catalyst) reached 98% conversion in 0.5 hours. - Example 8: 100 ml of NBR latex (no extra water) and 0.01253 g of HG2 catalyst were used to achieve direct latex hydrogenation.

Technical Relevance Provides an organic-solvent-free process for the direct selective hydrogenation of NBR latex using solid-state addition of metathesis catalysts, avoiding the gelation and cross-linking issues typical of diimide-based latex hydrogenation.

Patent 5

- Patent Number: EP4221260B1
- Patent Title: MEMBRAN AND SOUND GENERATION DEVICE
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2025
- Relevance: Describes a diaphragm made of a rubber compound comprising NBR and HNBR with specific acrylonitrile block contents for acoustic applications.

Polymerization / Synthesis Method - Polymer Blend: Nitrile rubber (NBR) and Hydrogenated nitrile rubber (HNBR) - Acrylonitrile Block Content: 20% to 80% by mass - Mass Ratio (NBR/HNBR): No less than 1 - Filler Content: 5% to 60% by mass (e.g., carbon black, white carbon black, calcium carbonate) - Vulcanizing Agent Content: 0.5% to 5% by mass - Operating Temperature: -80\u00b0C to 180\u00b0C

Relevant Experimental Evidence - The patent evaluates the effect of acrylonitrile block content (5 wt% to 85 wt%) on diaphragm performance. - At 20 wt% acrylonitrile, the glass transition temperature (Tg) is -31.1\u00b0C, tensile strength is 28.3 MPa, density is 0.96 g/cm\u00b3, and resilience is 85%. - At 50 wt% acrylonitrile, Tg is -23.5\u00b0C, tensile strength is 32.9 MPa, density is 1.11 g/cm\u00b3, and resilience is 81%. - At 80 wt% acrylonitrile, Tg is -17.1\u00b0C, tensile strength is 35 MPa, density is 1.19 g/cm\u00b3, and resilience is 78%. - Controlling the acrylonitrile content between 20% and 80% ensures an optimal balance of cold resistance (low Tg), mechanical strength, and resilience.

Technical Relevance Illustrates a practical application of HNBR/NBR blends in high-performance acoustic diaphragms, demonstrating how the acrylonitrile block content dictates the low-temperature flexibility, damping, and mechanical properties of the vulcanized rubber compound.

Patent 6

- Patent Number: EP3030824B1
- Patent Title: Method for the production of and high temperature field joints

- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2024
- Relevance: Mentions high-temperature resistant rubber mixtures (such as partially uncrosslinked rubber or fluoroelastomers) for pipeline insulation operating at 150\u00b0C to 205\u00b0C.

Polymerization / Synthesis Method - Operating Temperature: 150\u00b0C to above 205\u00b0C - Thermal Conductivity: Less than about 0.40 W/mk

Relevant Experimental Evidence - No specific experimental examples disclosed in the supplied evidence.

Technical Relevance Discusses the application of high-temperature resistant elastomeric materials (which can include HNBR and fluoroelastomers) in pipeline field joints, highlighting their thermal insulation and mechanical performance under extreme continuous operating temperatures.

Patent 7

- Patent Number: US11192293B2
- Patent Title: Process for producing 3D structures from rubber material
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2021
- Relevance: Describes a 3D printing process for rubber materials, specifically mentioning HNBR (Therban) as a suitable high-viscosity material.

Polymerization / Synthesis Method - Mooney Viscosity: >10 ME at 60\u00b0 C. and <200 ME at 100\u00b0 C. before curing - Ingredients: THERBAN AT 3404 (HNBR, 100 phr), PERBUNAN 2831 F (NBR, 100 phr), LEVAPREN 600 (EVM, 100 phr), BAYPREN 210 (CR, 100 phr), KELTAN 2470L (EPDM, 100 phr) - Filler: CORAX N 550/30 (30 phr) - Nozzle Geometry: L/D = 20 mm / 2 mm or L/D = 0.6 mm / 0.4 mm - Preheating: 100\u00b0 C. for 2 minutes - Curing Temperature: 180\u00b0 C. - Curing Time: 12 minutes (press-molded) or 15 minutes (atmospheric pressure)

Relevant Experimental Evidence - The patent provides examples of rubber formulations (1 *to 5*) using different base rubbers: 1 *uses THERBAN AT 3404 (HNBR), 2* uses PERBUNAN 2831 F (NBR), 3 *uses LEVAPREN 600, 4* uses BAYPREN 210, and 5 *uses KELTAN 2470L. - Each formulation contains 100 phr rubber, 30 phr CORAX N 550/30 carbon black, and other standard additives. - Filament forming was tested in a G\u00f6ttfert capillary viscosimeter at 100\u00b0C with a 2-minute preheating step. - The printed precursors were cured at 180\u00b0C for 15 minutes under atmospheric pressure, resulting in Shore A hardness values of 22 (for HNBR foam/structure 1*), 42 (2*), 35 (3*), 50 (4*), and 26 (5*).

Technical Relevance Demonstrates the feasibility of using HNBR (Therban) in additive manufacturing (3D printing) by utilizing its specific rheological window (Mooney viscosity >10 ME at 60\u00b0C and <200 ME at 100\u00b0C) to maintain shape stability before thermal curing.

Patent 8

- Patent Number: EP1825913B1
- Patent Title: New catalyst systems and their application for metathesis reactions
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2016
- Relevance: Describes a metathesis process for degrading NBR using a catalyst system comprising a metathesis catalyst and specific salts to facilitate subsequent hydrogenation to low-Mooney HNBR.

Polymerization / Synthesis Method - Starting Material: Nitrile rubber (NBR) - Catalyst: Ruthenium or Osmium-based metathesis catalyst (e.g., Grubbs I or Grubbs II) - Additive: Salt of formula K^n+ A^z- (e.g., lithium, sodium, potassium halides/sulfonates) - Ruthenium Content: 61 ppm to 614 ppm based on NBR

Relevant Experimental Evidence - The patent describes the metathesis degradation of NBR in the presence of a co-olefin and a catalyst system containing a transition metal complex (e.g., Grubbs I or II) and an inorganic or organic salt. - The addition of the salt improves the activity and stability of the metathesis catalyst, allowing lower catalyst loadings (e.g., reducing Ru content below 61 ppm based on NBR). - The degraded NBR is subsequently hydrogenated in-situ to yield HNBR with a weight average molecular weight (Mw) of 30,000 to 250,000, a Mooney viscosity (ML 1+4 @ 100\u00b0C) of 3 to 50, and a polydispersity index of less than 2.5.

Technical Relevance Contributes a highly active metathesis catalyst system modified with salts to efficiently degrade NBR prior to hydrogenation, enabling the cost-effective production of low-molecular-weight, low-Mooney HNBR.

Patent 9

- Patent Number: US11349176B2
- Patent Title: Separator for all-solid-state batteries, method for producing the same, and all …
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2022
- Relevance: Utilizes a hydrogenated rubber-based resin (such as HNBR) as a binder in a solid electrolyte separator layer to achieve high tensile strength and ion conductivity.

Polymerization / Synthesis Method - Solid Electrolyte: 10LiI-15LiBr-75 (3Li2S-P2S5) (0.400 g, particle diameter 2 \u03bcm, density 2.21 g/cm\u00b3) - Binder: Hydrogenated rubber-based resin (e.g., HNBR) - Binder Content (First Layer): 15% to 30% by volume - Binder Content (Second Layer): 0.1% to <15% by volume - Slurry Solid Content: 80% by mass or less - Compacting Pressure: 0.2 to 0.3 t/cm\u00b2 (\u224819.6 to 29.4 MPa) or up to 588 MPa - Pressing Temperature: 0\u00b0 C. or more (less than decomposition temp of resin)

Relevant Experimental Evidence - Example 1: A first solid electrolyte layer forming slurry was prepared using 0.400 g of 10LiI-15LiBr-75 sulfide solid electrolyte and a hydrogenated rubber-based resin binder. - The slurry was applied to a support and dried to form a precursor film, which was then pressed at a compacting pressure of 0.2 to 0.3 t/cm\u00b2 (\u224819.6 to 29.4 MPa) at 25\u00b0C. - Comparative Example 4: A solid electrolyte layer was pressed at 588 MPa for 30 minutes, pulverized, and re-pressed. - The use of 15-30 vol% hydrogenated rubber-based resin in the first layer and 0.1-<15 vol% in the second layer provides a separator with high tensile strength, excellent flexibility, and high lithium-ion conductivity.

Technical Relevance Demonstrates a novel electrochemical application of HNBR as a highly flexible, high-elongation polymeric binder in sulfide-based solid-state battery separators, maintaining excellent mechanical integrity without compromising ionic transport.

Patent 10

- Patent Number: US20230015442A1
- Patent Title: Method for separating and recovering lignin and meltable flowable biolignin …
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2023
- Relevance: Mentions blending lignin with nitrile rubber or HNBR to act as a filler/modifier.

Polymerization / Synthesis Method - Polymer Blend: Lignin and nitrile rubber (NBR) or acrylonitrile butadiene - Lignin Form: Dry lignin powder (used as a nano-filler) - Processing Temperature: Greater than 120\u00b0 C. (up to 180\u00b0 C. for glass transition)

Relevant Experimental Evidence - The patent discusses prior art (e.g., U.S. Pat. No. 9,453,129) where dry lignin powder is blended with nitrile rubber or acrylonitrile butadiene to act as a nano-filler. - It notes that such conventional attempts result in stiff and brittle composites due to the poor flowability and high glass transition temperature (169\u00b0C to 180\u00b0C) of the lignin powder. - No specific experimental examples of HNBR synthesis are disclosed in the supplied text, as the patent primarily focuses on methods for recovering meltable, flowable biolignin from biomass.

Technical Relevance Highlights the historical use and limitations of blending lignin with nitrile rubbers/HNBR, noting that conventional dry powder blending acts merely as a stiffening mineral filler due to poor melt flow characteristics.

Patent 11

- Patent Number: US10869391B2
- Patent Title: Garment-type electronic device and method for producing same
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2020
- Relevance: Mentions using synthetic rubber (which can include HNBR) as a stretchable binder resin for conductive pastes in wearable electronics.

Polymerization / Synthesis Method - Binder: Synthetic rubber, natural rubber, or polyurethane resin - Conductive Filler: Silver particles, carbon nanotubes, or silver fillers - Curing Conditions: Room temperature over 24 hours (for silicone resin layer) - Stretching: 10% strain evaluation

Relevant Experimental Evidence - Example 5 & 120: A two-component curable silicone resin was poured into a screen to a thickness of 5 mm and cured at room temperature over 24 hours. - Example 1: Evaluated the level difference at the boundary between the electrode portion and the wiring portion, and evaluated the maintenance of electrical conductivity under 10% stretching. - The conductive paste utilizes a stretchable binder resin (such as synthetic rubber) kneaded with conductive particles to form a printable paste. - The resulting printed wiring maintains electrical conductivity when deformed because the rubbery binder portion absorbs the external strain.

Technical Relevance Illustrates the application of highly stretchable synthetic rubbers (such as HNBR) as binder matrices in conductive inks for printed, wearable electronics, ensuring conductivity is maintained under repeated mechanical stretching.

Patent 12

- Patent Number: US10415370B2
- Patent Title: Systems and methods for in situ monitoring of cement slurry locations and …
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2019
- Relevance: Mentions that cement compositions can include polymers like epoxies and latexes (which can include nitrile rubber latexes) in place of or in addition to hydraulic cement.

Polymerization / Synthesis Method - Polymer Additives: Epoxies and latexes (used in cement fluid compositions)

Relevant Experimental Evidence - No specific experimental examples disclosed in the supplied evidence.

Technical Relevance Mentions the use of polymer latexes (such as nitrile rubber latexes) as modifiers in cement fluid compositions for downhole oil and gas applications, which are monitored in situ using optical analysis devices.

Patent 13

- Patent Number: US8536390B2
- Patent Title: Profitable method for carbon capture and storage
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2013
- Relevance: Describes producing bio-derived olefins (including butadiene, a key monomer for NBR/HNBR) from biomass via gasification, Fischer-Tropsch conversion, and steam cracking.

Polymerization / Synthesis Method - Fischer-Tropsch Catalyst: Cobalt-based catalyst supported on alumina, silica, titania, or zirconia, modified with Ru, Rh, or Pd, and promoted with Mg, K, or La - FT Hydrocarbon Range: n = 2 to 100 (mostly C5 to C50) - Steam Cracking Products: Ethylene, propylene, butenes, and butadiene

Relevant Experimental Evidence - The patent describes converting biomass (e.g., wood chips, agricultural residues, or algae) into paraffinic hydrocarbons via gasification and Fischer-Tropsch synthesis using a cobalt catalyst modified with Ru, Rh, or Pd. - The paraffinic hydrocarbons are steam cracked to produce bio-derived light olefins, specifically ethylene, propylene, butenes, and butadiene. - These bio-derived monomers can then be polymerized into non-biodegradable plastics, effectively sequestering atmospheric CO2. - No specific experimental examples of NBR or HNBR polymerization are disclosed in the supplied text.

Technical Relevance Provides a synthetic pathway for producing bio-derived butadiene, which is a critical monomer for the synthesis of environmentally sustainable nitrile rubber (NBR) and subsequent hydrogenated nitrile rubber (HNBR).

Patent 14

- Patent Number: US8247063B2
- Patent Title: Multilayer material and related methods
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2012
- Relevance: Mentions polyolefins, diene copolymers, and rubber-based hot melt adhesives in multilayer packaging materials.

Polymerization / Synthesis Method - Stretching/Drawing Ratio: 1.1:1 to 9:1 (preferably 4.6-5.4:1 or 5.5-6.5:1) - Stretching Temperature: 90\u00b0C to 140\u00b0C (preferably 108-118\u00b0C or 116-126\u00b0C) - Adhesive Coating Weight: 10 to 50 g/m\u00b2

Relevant Experimental Evidence - The patent describes a multilayer material comprising a first printable transparent layer (e.g., PET, BOPP, LLDPE) and a second layer (e.g., foil) coupled via an adhesive interface. - The film is oriented in the machine direction at a draw ratio of 1.1:1 to 9:1

at a temperature of 90\u00b0C to 140\u00b0C. - Adhesives used at the interface include acrylic or rubber-based hot melt adhesives, solvent acrylics, or urethanes applied at 10 to 50 gsm. - No specific experimental examples of HNBR synthesis are disclosed in the supplied text.

Technical Relevance Mentions the use of rubber-based adhesives and elastomeric polyolefin layers in coextruded multilayer films, representing potential application areas for flexible HNBR-based adhesive formulations.

Patent 15

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2017
- Relevance: Describes preparing meta-block copolymers by contacting a metathesis catalyst with ethylenically unsaturated polymer reagents, specifically exemplifying a blend of partially-hydrogenated polybutadiene (HPBD) and partially-hydrogenated nitrile/butadiene rubber (HNBR).

Polymerization / Synthesis Method - Polymer Reagents: Partially-hydrogenated polybutadiene (HPBD, 0.25 g) and partially-hydrogenated nitrile/butadiene rubber (HNBR, 0.25 g) - Solvent: Toluene - Reaction Temperature: 95\u00b0 C. or 105\u00b0 C. - Catalyst: Grubbs II metathesis catalyst (8 mg or 9 mg) - Product Recovery: 0.45 g, 0.47 g, or 0.71 g under reduced pressure - Extraction Solvent: Tetrahydrofuran (30 ml or 50 ml)

Relevant Experimental Evidence - Example 1: A toluene solution containing 0.25 g each of HPBD and HNBR was warmed to 95\u00b0C. 8 mg of Grubbs II metathesis catalyst was added. Removing volatile components under reduced pressure yielded 0.45 g of recovered meta-block copolymer product. - Example 6: A toluene solution containing 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-octene-co-butadiene) was warmed to 105\u00b0C, and 9 mg of Grubbs II catalyst was added, yielding 0.47 g of polymer product showing microphase order at 300\u00b0C. - Example 7: A toluene solution containing 0.10 g poly(ethylene-co-butadiene) and 0.40 g poly(ethylene-co-octene-co-butadiene) was reacted at 105\u00b0C, yielding 0.47 g of recovered polymer product analyzed by CRYSTAF. - Example 8: Recovered product was washed with 30 ml THF overnight to extract non-metathesized unsaturated poly(ethylene oxide), showing microphase order at 280\u00b0C. - Example 10: Recovered 0.71 g of polymer product, which was washed with 50 ml THF overnight to extract non-metathesized components.

Technical Relevance Demonstrates the synthesis of novel meta-block copolymers containing HNBR segments via transition-metal-catalyzed polymer segment interchange (cross-metathesis), providing a route to block copolymers with tailored elastomeric and polar properties.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Polymerization and Modification Approaches: Direct emulsion polymerization of NBR followed by homogeneous solution hydrogenation (US20210340285A1, US20160326272A1) is compared against direct latex-phase hydrogenation without organic solvents (US9657113B2), which avoids the environmental footprint and gelation issues of solution-based methods.
- Molecular Weight Control: Use of olefin metathesis (Grubbs catalysts) to degrade NBR chains prior to hydrogenation to achieve low-Mooney HNBR (US7919563B2, EP1825913B1) is contrasted with segment interchange metathesis to form meta-block copolymers containing HNBR blocks (EP2057205B1).
- Catalyst Systems and Cocatalysts: Homogeneous systems utilizing Ru, Pd, or Rh catalysts with triphenylphosphine (TPP) cocatalysts (US20160326272A1) are compared with low-concentration catalyst systems designed to remain in the polymer without affecting ageing (US20210340285A1).
- Formulation and Application Trends: Diverse applications are explored, including blending HNBR with NBR for acoustic diaphragms (EP4221260B1), using HNBR as a high-elongation binder in solid-state battery separators (US11349176B2), and utilizing HNBR in stretchable conductive inks for wearable electronics (US10869391B2).

4. CONCLUSION

In conclusion, the synthesis of high-performance Hydrogenated Nitrile Butadiene Rubber (HNBR) relies on precise control over hydrogenation chemistry, molecular weight distribution, and residual catalyst/cocatalyst chemistry. For applications requiring standard high-molecular-weight HNBR, solution hydrogenation using low-concentration ruthenium or palladium catalysts (42–79 ppm Ru or 44–67 ppm Pd) at 60–200°C and 0.7–15 MPa is recommended to eliminate the need for catalyst recovery while maintaining excellent thermal ageing resistance. When using homogeneous catalysts with triphenylphosphine (TPP) cocatalysts, a post-treatment oxidation step using mild oxidizing agents (e.g., potassium chlorate or peroxides) is highly recommended to convert residual TPP to phosphine oxides, thereby preventing modulus degradation and high compression set in the vulcanizates. For low-viscosity processing, a two-step metathesis-hydrogenation process using 0.05–0.25 phr Grubbs catalyst in monochlorobenzene with an ethylene or 1-hexene chain transfer agent at 80°C successfully yields low-Mooney HNBR (ML 1+4 @ 100°C of 3–50) with narrow polydispersity (<2.5). Finally, direct latex-phase hydrogenation using solid-state Hoveyda-Grubbs catalysts at 100–120°C and 3.45–6.89 MPa offers a promising, solvent-free alternative that avoids the environmental footprint of organic solvents.

5. REFERENCES

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- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed in the available patent text. | US | 2011 | https://patents.google.com/patent/US7919563B2/en
- US9657113B2 | Hydrogenation of a diene-based polymer latex | Not disclosed in the available patent text. | US | 2017 | https://patents.google.com/patent/US9657113B2/en
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- EP3030824B1 | Method for the production of and high temperature field joints | Not disclosed in the available patent text. | EP | 2024 | https://patents.google.com/patent/EP3030824B1/en
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- EP1825913B1 | New catalyst systems and their application for metathesis reactions | Not disclosed in the available patent text. | EP | 2016 | https://patents.google.com/patent/EP1825913B1/en
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- US20230015442A1 | Method for separating and recovering lignin and meltable flowable biolignin … | Not disclosed in the available patent text. | US | 2023 | https://patents.google.com/patent/US20230015442A1/en
- US10869391B2 | Garment-type electronic device and method for producing same | Not disclosed in the available patent text. | US | 2020 | https://patents.google.com/patent/US10869391B2/en
- US10415370B2 | Systems and methods for in situ monitoring of cement slurry locations and … | Not disclosed in the available patent text. | US | 2019 | https://patents.google.com/patent/US10415370B2/en
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